

Materials Science 1 - Failure of materials

Failure of materials can cause disasters



Failure of materials

Usually, the causes of accidents include inappropriate selection of materials and design of components, as well as inappropriate maintenance.

It is the responsibility of the engineers to anticipate and prevent failures, by an appropriate assessment of materials and structures.

This can be only achieved by a deep understanding of the mechanisms of failure of materials and strategies to reduce the risks.

In this instance, we will look at three main aspects of failure:

1. Fracture;
2. Fatigue;
3. Creep.

Learning outcomes of this lecture

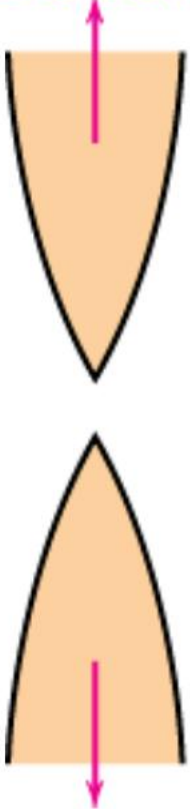
- Differentiate between crack propagation behaviour in ductile and brittle materials.
- Analysis of crack propagation using fractography.
- Identify intergranular and transgranular crack propagation modes.
- Distinguish between theoretical and real behaviour of failure.
- Understand the concept of stress intensity factor and fracture toughness.
- Calculate stress in materials containing cracks.
- Apply fracture mechanics in design of components.

Generalities of fracture behaviour

- Simple fracture is the separation of a body into two or more pieces in response to an applied stress that can be static or changing over time.
- Applied stresses causing fracture can be may be tensile, compressive, shear, torsional or combinations of them.
- For metals, two main fracture modes can be distinguished: **ductile and brittle**.
- Ductile and brittle are relative terms; whether a materials experiences ductile or brittle failure depends on various factors (i.e. temperature, loading conditions, loading rate, etc).

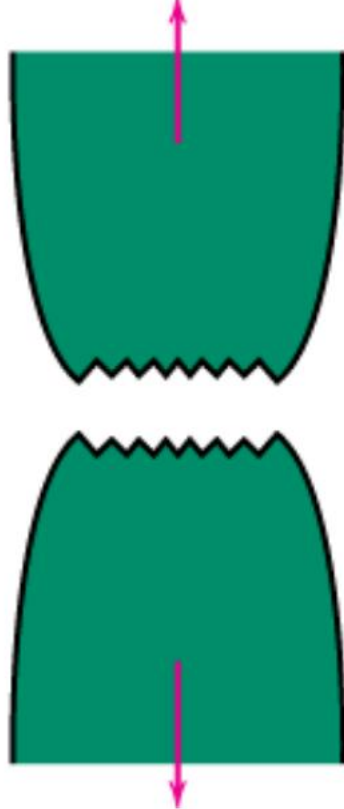
General characteristics of ductile and brittle fracture

Very
Ductile



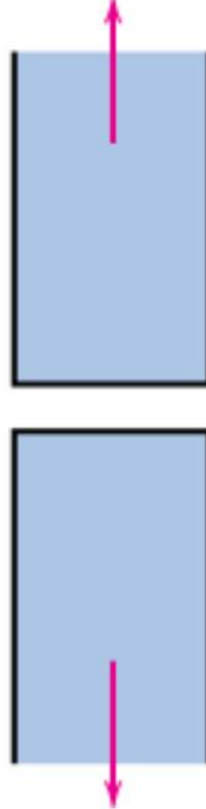
Highly soft metal, pure gold and lead at room T. Metals, polymers, and inorganic glasses at elevated temperatures.

Moderately
Ductile



Most common ductile fracture mode.

Brittle



Most ceramics and brittle metals have a brittle fracture mode.

Ductile fracture

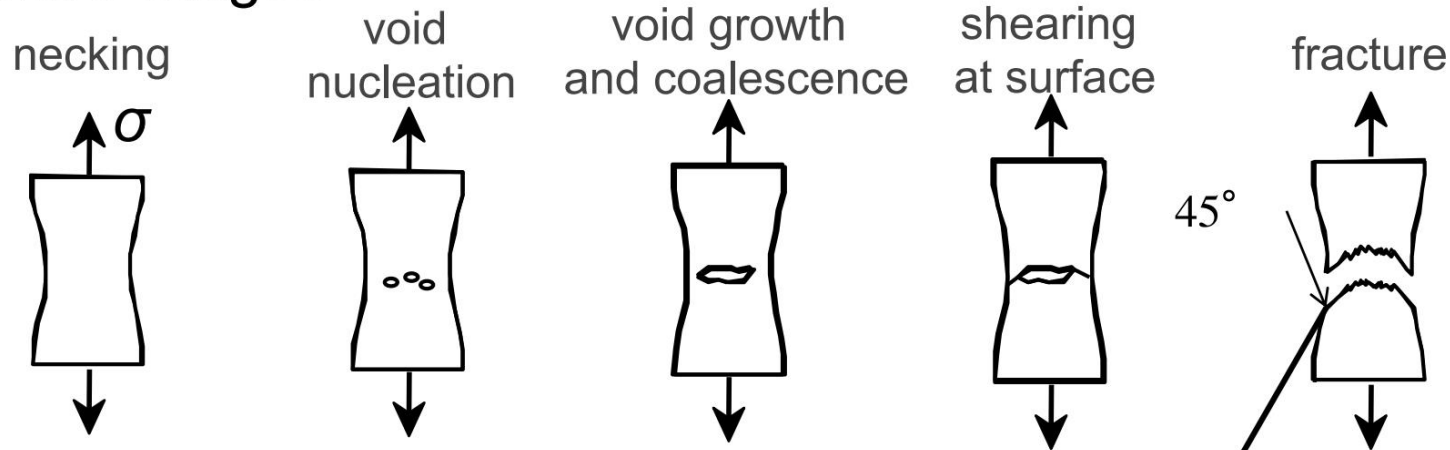
- Accompanied by significant plastic deformation (large %EL and large %AR);
- Proceeds relatively slowly, it needs additional energy to propagate;
- Plastic deformations represent warning signs of imminent failure.

Brittle fracture

- Accompanied by very limited or no plastic deformations.
- It occurs very rapidly; once started, it will continue spontaneously without an increase in the applied stress;
- It is catastrophic and often brittle components can shatter in many pieces.

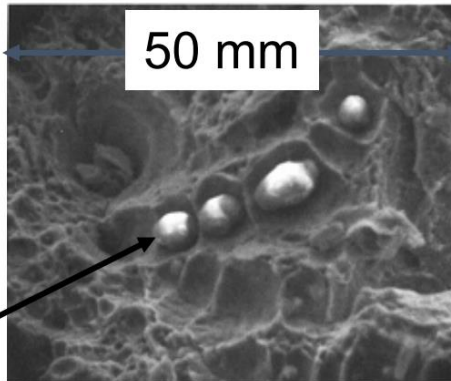
Moderately ductile fracture process

Failure Stages

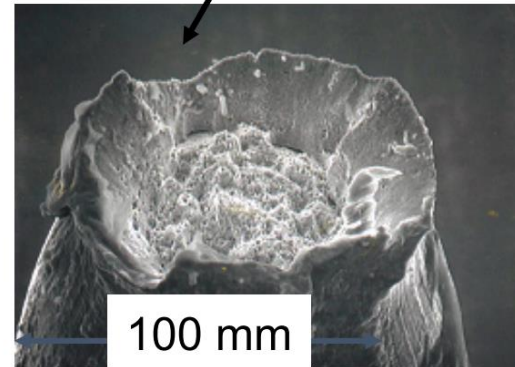


Resulting Fracture Surfaces (steel)

particles serve as void nucleation sites.



From V.J. Colangelo and F.A. Heiser, *Analysis of Metallurgical Failures* (2nd ed.), Fig. 11.28, p. 294, John Wiley and Sons, Inc., 1987. (Orig. source: P. Thornton, *J. Mater. Sci.*, Vol. 6, 1971, pp. 347-56.)



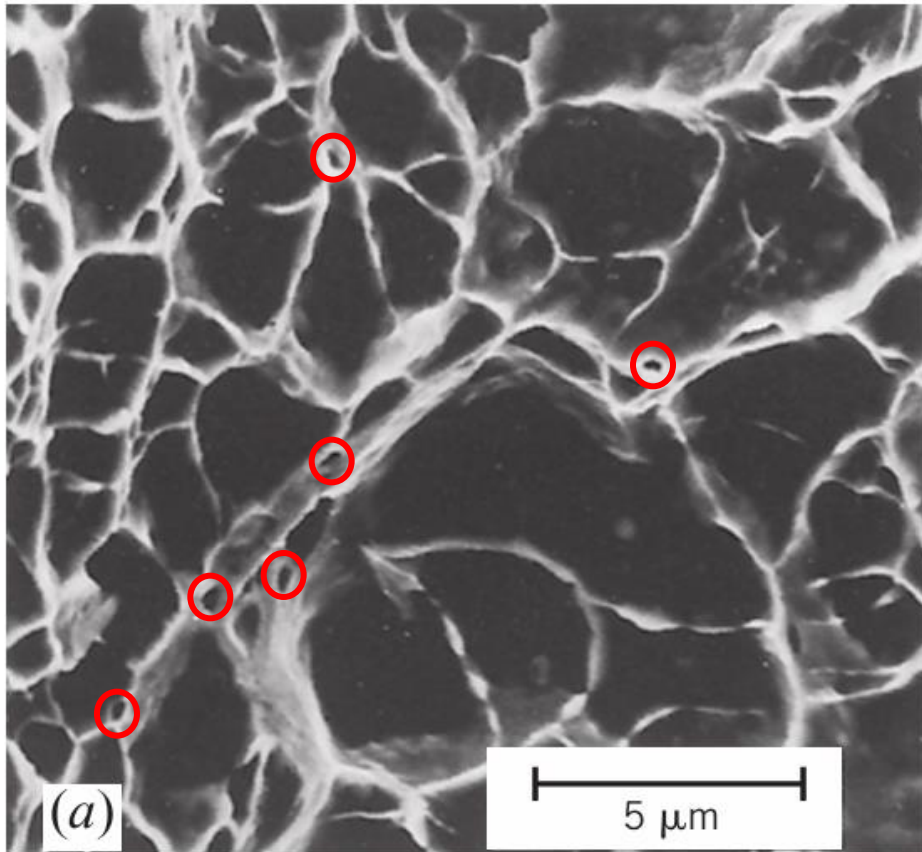
Fracture surface of tire cord wire loaded in tension. Courtesy of F. Roehrig, CC Technologies, Dublin, OH. Used with permission.

After void growth and coalescence, the fracture process proceeds by the rapid propagation of a crack around the outer perimeter of the neck, by shear deformation at about 45° with the tensile axis.

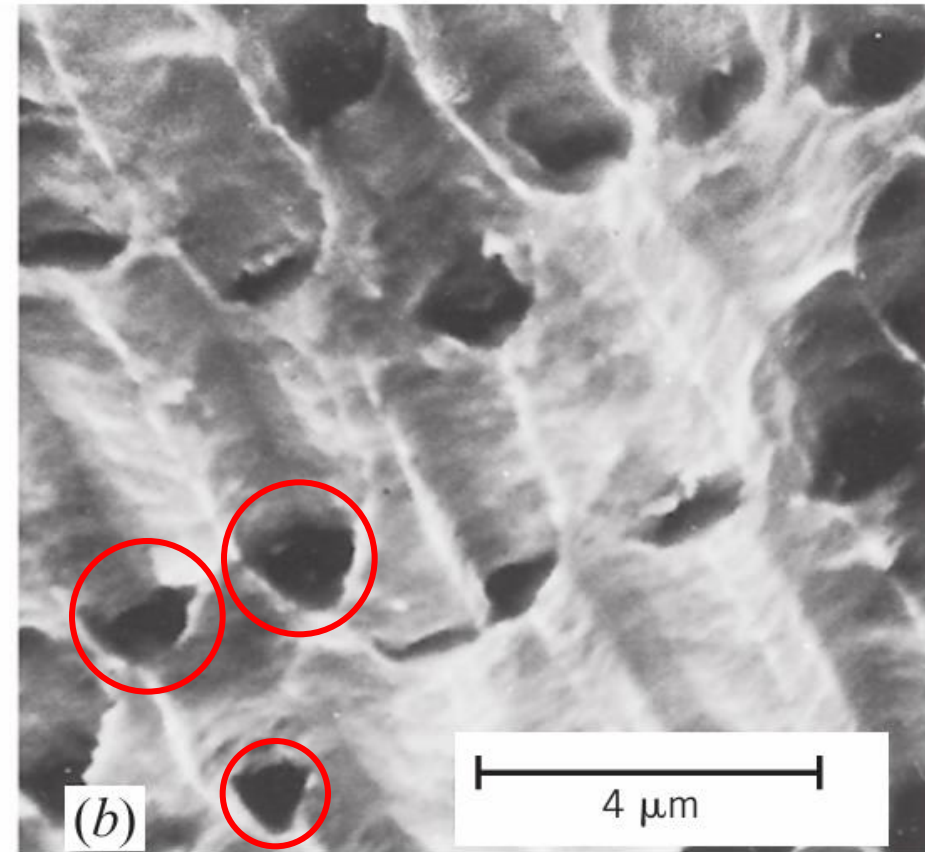
The fracture surface is often defined as cup-and-cone fracture.

Moderately ductile fracture process

Detailed information regarding the fracture mechanisms can be obtained using scanning electron microscopy (higher resolution than optical). Studies of this type are termed **fractographic**.



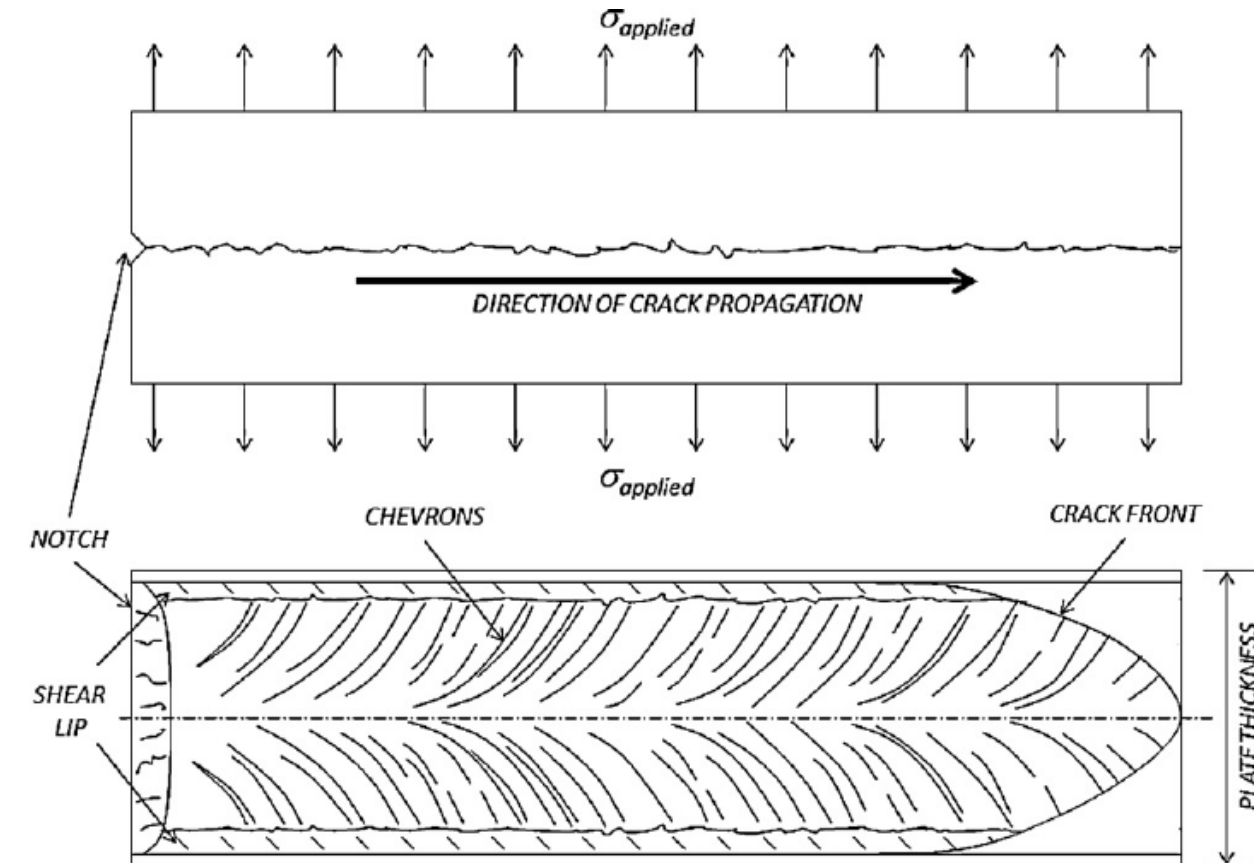
Scanning electron micrograph showing spherical-like dimples from uniaxial tensile loads.



Scanning electron micrograph showing parabolic-like shaped dimples resulting from shear loading.

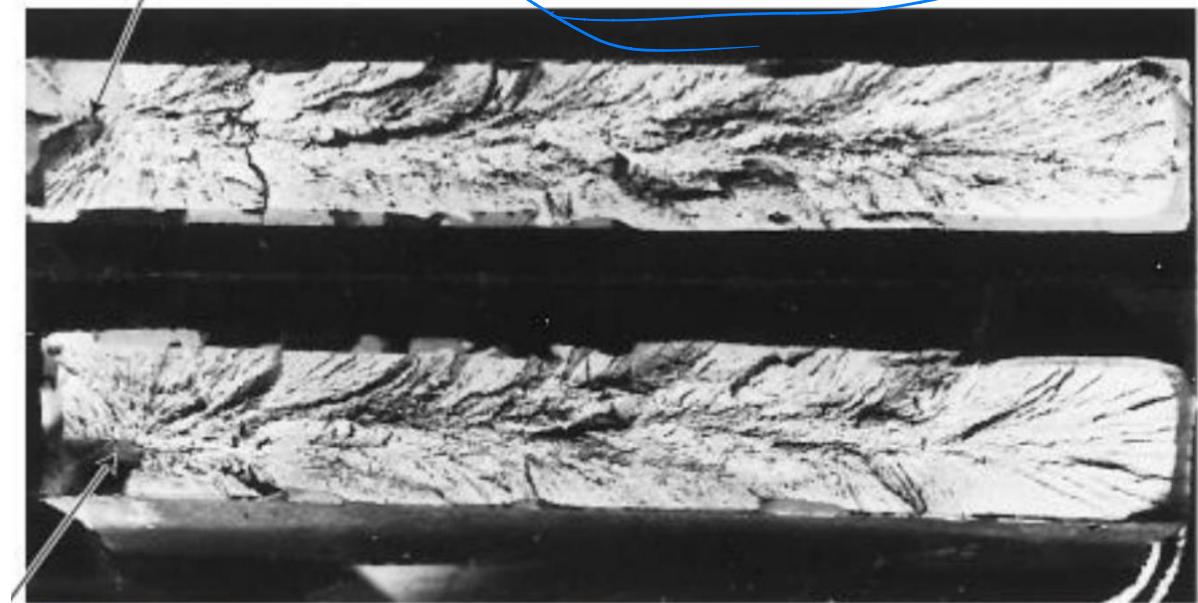
Brittle fracture

Brittle fracture takes place without any appreciable deformation and by rapid crack propagation. The direction of crack motion is nearly perpendicular to the direction of the applied tensile stress and yields a relatively flat fracture surface. Usually, the original shape of fractured pieces can be recombined.



V-shaped Chevron markings.
Also known as herringbone markings

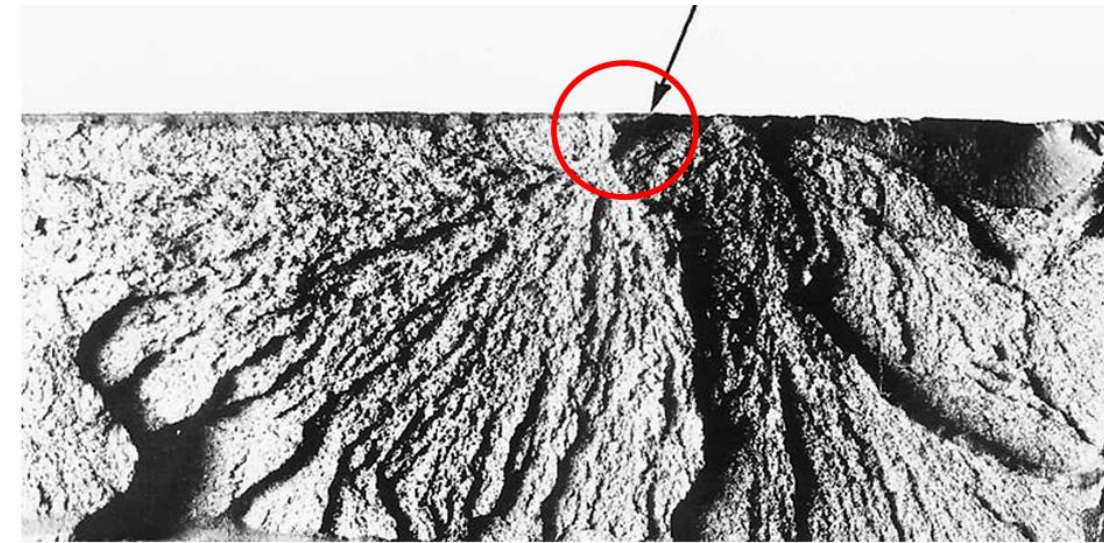
Arrows indicate point at failure origination



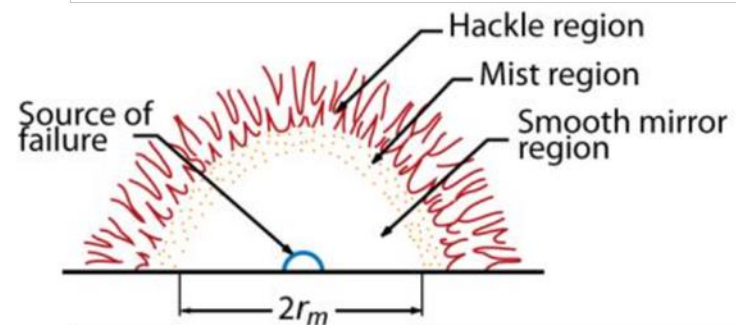
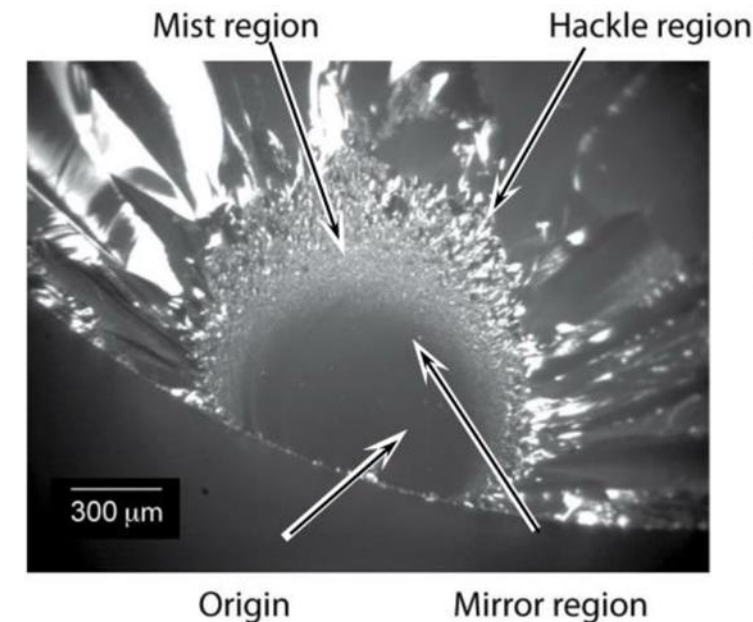
V-shaped Chevron markings in a steel sample. The Chevron markings usually point to the failure origin. Crack propagation may be discontinuous depending on the brittleness of the material.

Brittle fracture

Another typical brittle fracture feature is represented by fan-like fracture pattern.



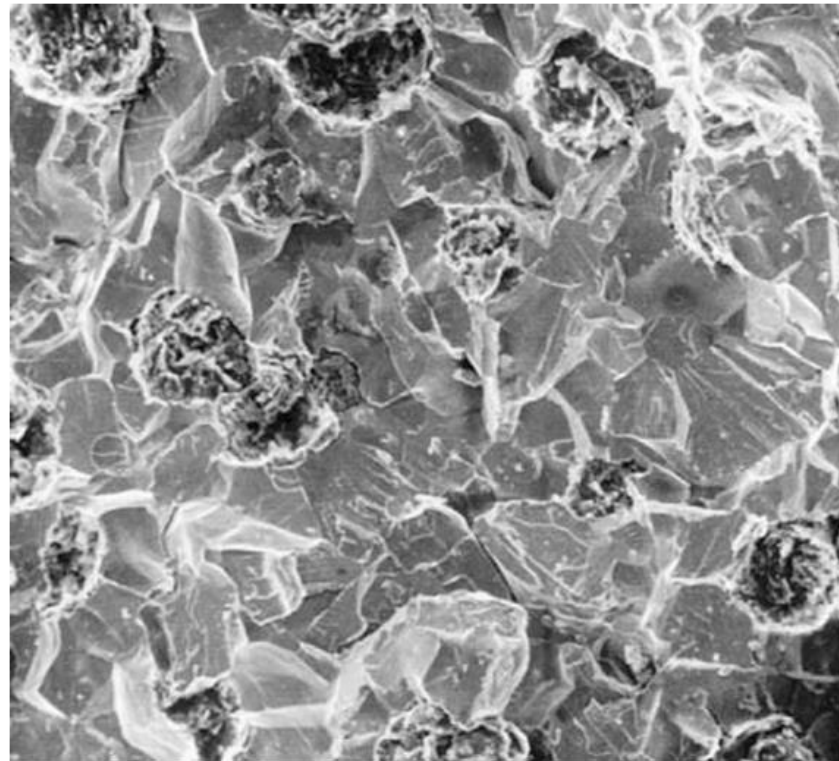
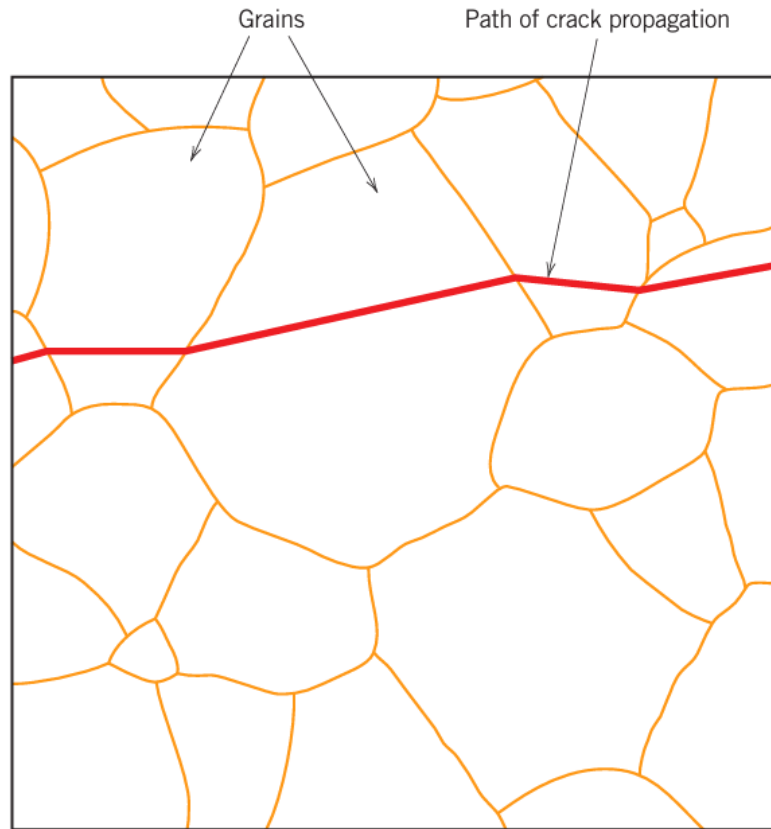
In metals, lines or ridges radiate from the origin of the crack in a fanlike pattern.



For ceramics and glasses different regions can be commonly observed on the fracture surface. The image refers to a silica rod.

Brittle fracture – Transgranular fracture

For most brittle crystalline materials, crack propagation corresponds to the successive and repeated breaking of atomic bonds along specific crystallographic planes called **cleavage planes**. This type of fracture is named **transgranular**, because the cracks pass through the grains.



SEM fracture surface of ductile cast iron

Cleavage planes represent planes of weak bonding in the crystal structure.

The fracture surface appears as a grainy or faceted texture as a result of changes in orientation of the cleavage planes from grain to grain.

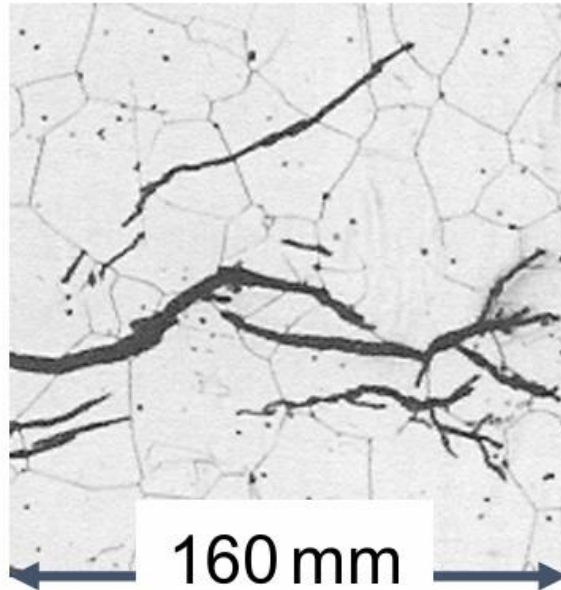
Brittle fracture – Transgranular fracture

Transgranular fracture can take place both in metals and in ceramics.

316 Stainless Steel (metal)

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from "Metals Handbook",
9th ed, Fig. 650, p. 357.
Copyright 1985, ASM
International, Materials
Park, OH. (Micrograph by
D.R. Diercks, Argonne
National Lab.)

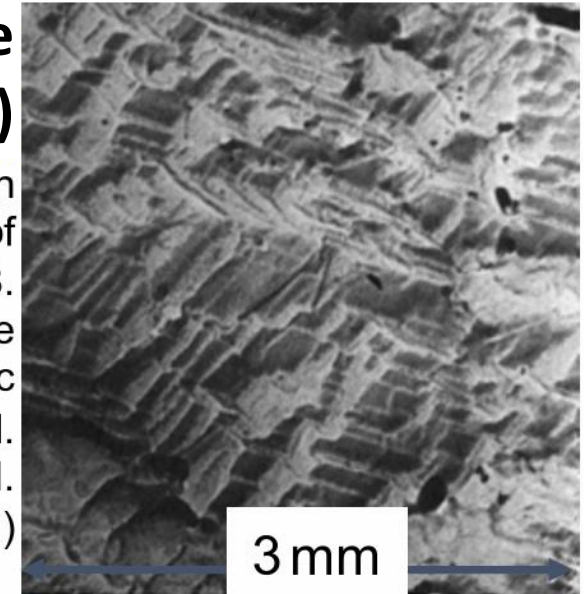
Polished and etched surface



Aluminium oxide (ceramic)

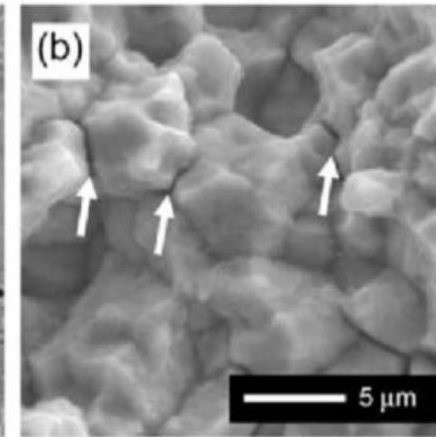
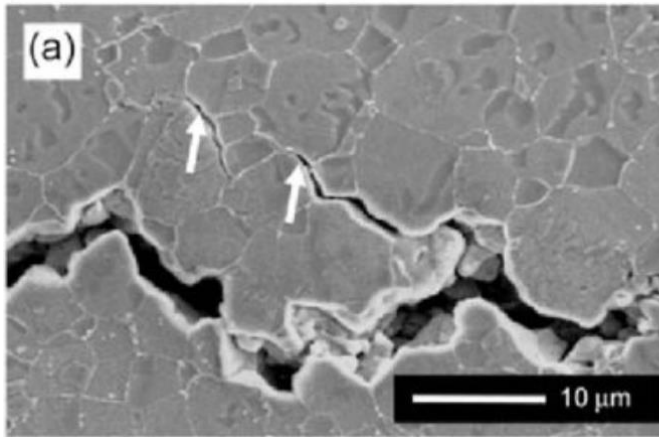
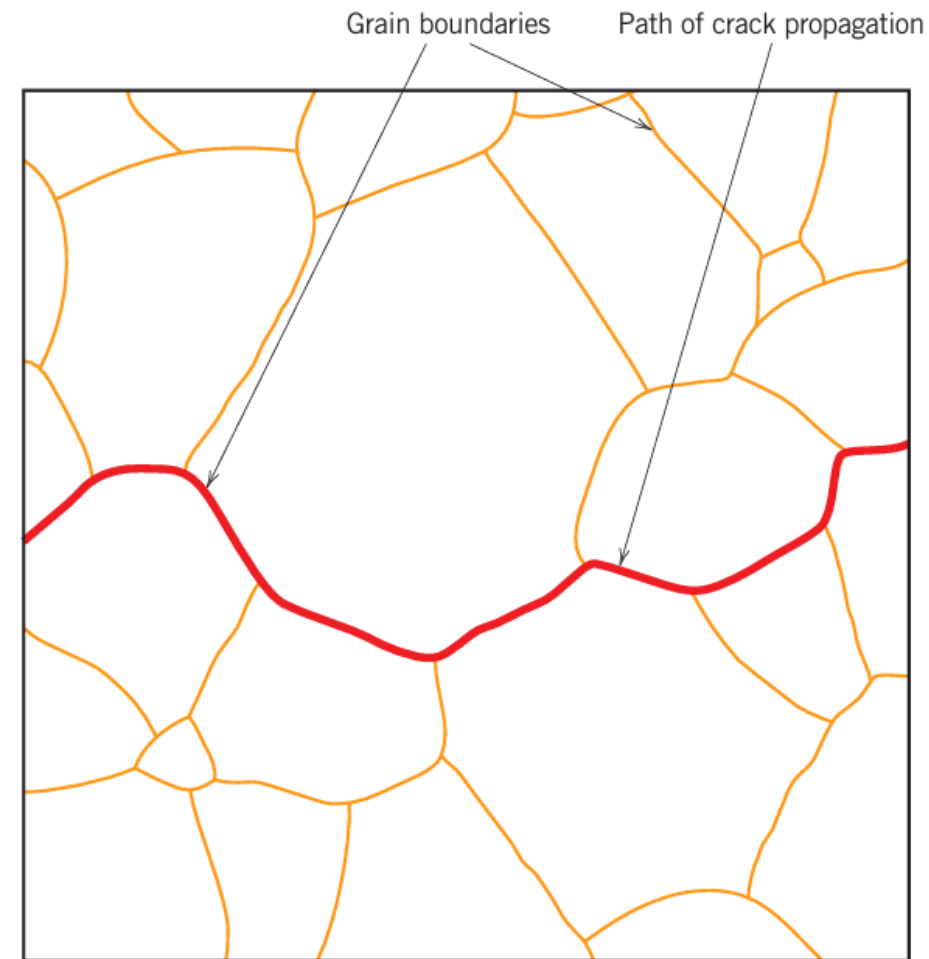
Reprinted w/ permission
from "Failure Analysis of
Brittle Materials", p. 78.
Copyright 1990, The
American Ceramic
Society, Westerville, OH.
(Micrograph by R.M.
Gruver and H. Kirchner.)

Fracture surface

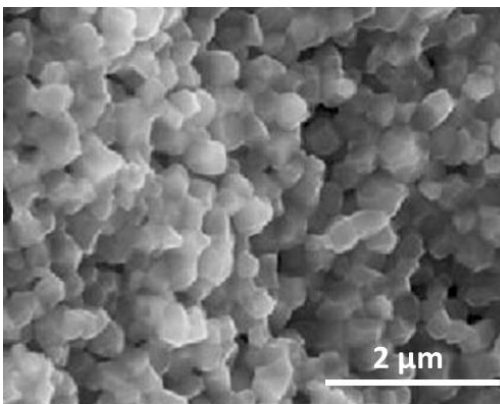


Brittle fracture – Intergranular fracture

The other typical fracture mode is termed **intergranular fracture**, and it is characterized by crack propagation along the grain boundaries. This type of fracture is usually caused by processes that weaken grain boundary regions.



(a) crack path and
(b) fracture surface
in nickel superalloy.



Fracture surface of
bismuth-sodium titanate
ceramics.

In the fracture surface, intergranular fracture, can be distinguished by features that resemble the three-dimensional nature of the grains.

Principles of fracture mechanics

Theoretical strength: stress required to cause fracture in an ideal single crystal across a particular crystallographic plane. It can be estimated from the following relationship:

$$\sigma_f = \left(\frac{E\gamma}{a_0} \right)^{\frac{1}{2}}$$

- E : Young's modulus.
- γ : energy to create new surface.
- a_0 : interatomic equilibrium distance.

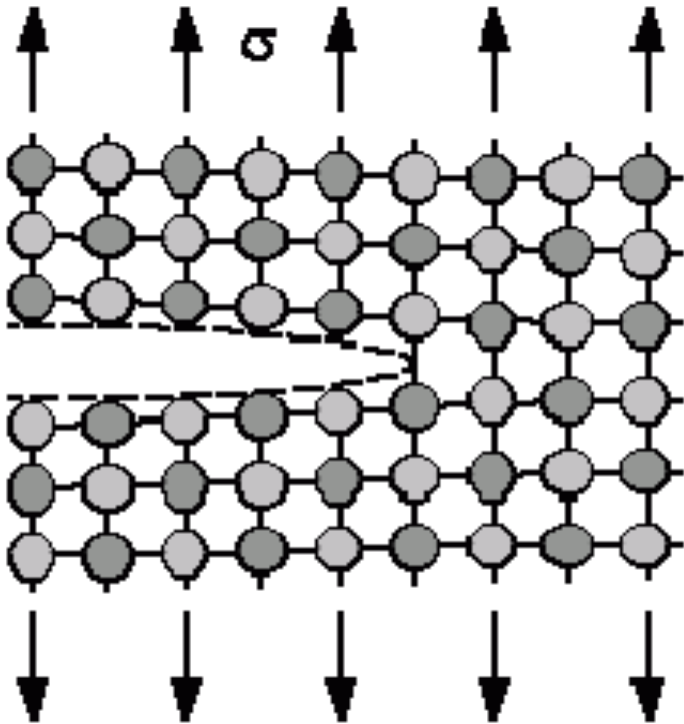
Significant discrepancy between the value predicted with previous equation and the experimental measured values.

Theoretical predictions

$$\frac{E}{10} \leq \sigma_f \leq \frac{E}{5}$$

Experimentally measured

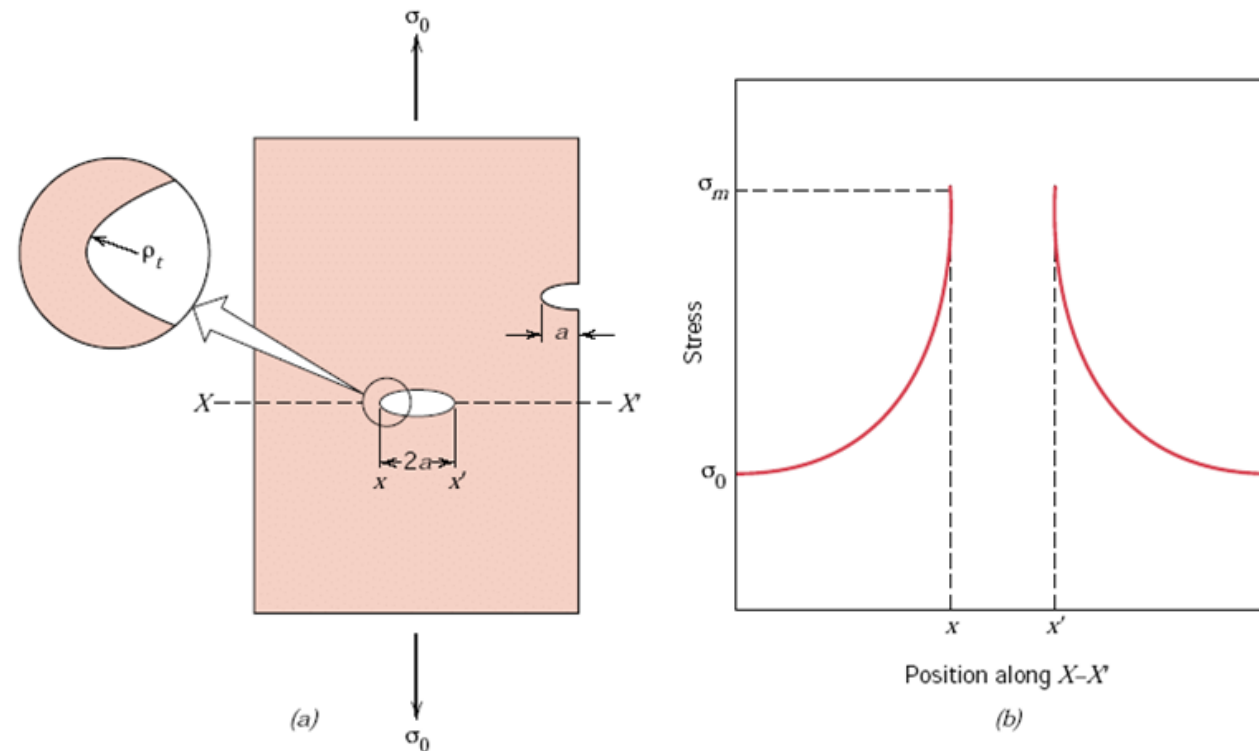
$$\sigma_f \approx \frac{E}{1000}$$



Principles of fracture mechanics

The measured fracture strength for most materials is significantly lower than that predicted by theoretical calculations based on atomic bonding energies. This is due to the presence of microscopic flaws or cracks that always exist in the material.

Under applied forces, the stress is amplified in the vicinity of defect/crack tips. This is quantified by the **stress intensity factor** or **stress concentration factor**.



$$\sigma_m = 2\sigma_o \left(\frac{a}{\rho_t} \right)^{1/2} = K_t \sigma_o$$

where

a = one half the length of an internal crack

ρ_t = radius of curvature of the crack tip

σ_o = applied stress

σ_m = stress at crack tip

K_t = stress concentration factor

Principles of fracture mechanics

Example 1:

What is the magnitude of the maximum stress that occurs at the tip of an internal crack having a radius of curvature of 2.5×10^{-4} mm and a crack length of 2.5×10^{-2} mm when a tensile stress of 170 MPa is applied?

$$\sigma_m = 2 \sqrt{\frac{a}{r}} \sigma_t$$

$$= 2 \sqrt{\frac{\frac{5}{4} \times 10^{-5}}{\frac{2.5 \times 10^{-7}}{2}}}$$
$$= 2 \sqrt{\frac{1}{2} \times 100}$$

$$170 \times 10^3$$

$$\approx 2 \sqrt{50} \times 170 \times 10^3 \text{ Pa}$$
$$= 2 \sqrt{50} \times 170 \text{ MPa}$$

Principles of fracture mechanics

Example 1 solution:

What is the magnitude of the maximum stress that occurs at the tip of an internal crack having a radius of curvature of 2.5×10^{-4} mm and a crack length of 2.5×10^{-2} mm when a tensile stress of 170 MPa is applied?

$$\sigma_m = 2\sigma_o \left(\frac{a}{\rho_t} \right)^{1/2} = K_t \sigma_o$$

$$\sigma_m = 2(170 \text{ MPa}) \left(\frac{1.25 \times 10^{-2} \text{ mm}}{2.5 \times 10^{-4} \text{ mm}} \right)^{1/2}$$

$$\sigma_m = 2404 \text{ MPa}$$

Principles of fracture mechanics

Example 2:

Estimate the maximum stress on cracks produced on a brittle material if it is known that fracture occurs by the propagation of an elliptically shaped surface crack of length 0.5 mm and a tip radius of curvature of 5×10^{-3} mm, when a stress of 1035 MPa is applied.

$$G_m = 260 \left(\frac{a}{\rho_t} \right)^{\frac{1}{2}}$$

$$= 2 \sqrt{\frac{0.5 \times 10^{-3}}{2}} \sqrt{\frac{1035 \text{ MPa}}{5 \times 10^{-6}}}$$

$$1035 \text{ MPa}$$

=

Principles of fracture mechanics

Example 2 solution:

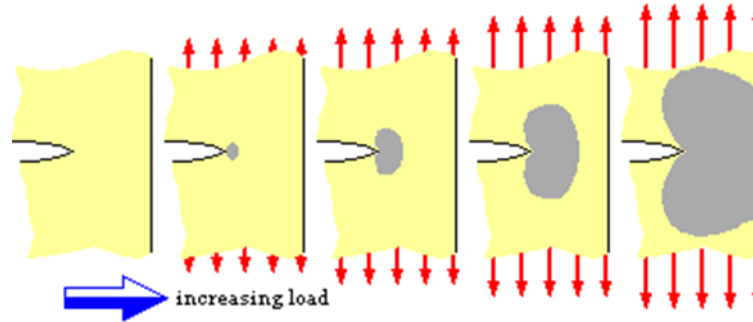
Estimate the maximum stress on cracks produced on a brittle material if it is known that fracture occurs by the propagation of an elliptically shaped surface crack of length 0.5 mm and a tip radius of curvature of 5×10^{-3} mm, when a stress of 1035 MPa is applied.

Calculate σ_m given that $\sigma_0 = 1035$ MPa, $a = 0.5$ mm, and $\rho_t = 5 \times 10^{-3}$ mm

$$\begin{aligned}\sigma_m &= 2\sigma_0 \left(\frac{a}{\rho_t} \right)^{1/2} \\ &= (2)(1035 \text{ MPa}) \left(\frac{0.5 \text{ mm}}{5 \times 10^{-3} \text{ mm}} \right)^{1/2} \\ &= 2.07 \times 10^4 \text{ MPa} = 20.7 \text{ GPa}\end{aligned}$$

Principles of fracture mechanics

The effect of a stress raiser is more significant in brittle than in ductile materials. In ductile materials, when the stress at the tip is larger than the yield stress, plastic deformations are developed around the crack tip, hindering crack propagation.



In brittle materials, yielding and stress redistribution around the crack tip do not occur. Using principles of fracture mechanics, it can be demonstrated that the **critical stress** σ_c required for crack propagation in a brittle material can be estimated by the expression:

$$\sigma_c = \left(\frac{2E\gamma_s}{\pi a} \right)^{1/2}$$

- E = modulus of elasticity
- γ_s = specific surface energy
- a = one half length of internal crack

Crack propagation occurs when $\sigma_m > \sigma_c$

Principles of fracture mechanics

Example 3:

If the specific surface energy for aluminum oxide is 0.90 J/m^2 , calculate the critical stress required for the propagation of an internal crack of length 0.40 mm . The modulus of elasticity for this material is 393 GPa .

Principles of fracture mechanics

Example 3 solution:

If the specific surface energy for aluminium oxide is 0.90 J/m², calculate the critical stress required for the propagation of an internal crack of length 0.40 mm. The modulus of elasticity for this material is 393 GPa.

$$\sigma_c = \left(\frac{2E\gamma_s}{\pi a} \right)^{1/2} = \left[\frac{(2)(393 \times 10^9 \text{ N/m}^2)(0.90 \text{ N/m})}{(\pi) \left(\frac{0.4 \times 10^{-3} \text{ m}}{2} \right)} \right]^{1/2}$$
$$= 33.6 \times 10^6 \text{ N/m}^2 = 33.6 \text{ MPa}$$

Principles of fracture mechanics

Example 4:

An MgO component must not fail when a tensile stress of 13.5 MPa is applied. Determine the maximum allowable surface crack length if the surface energy of MgO is 1.0 J/m². The value for modulus of elasticity is 225 GPa.

Principles of fracture mechanics

Example 4 solution:

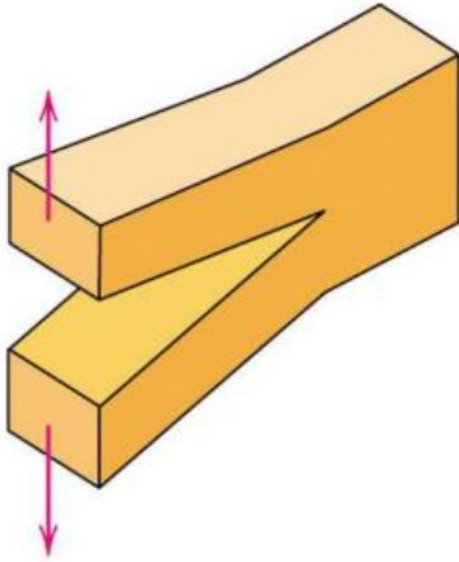
An MgO component must not fail when a tensile stress of 13.5 MPa is applied. Determine the maximum allowable surface crack length if the surface energy of MgO is 1.0 J/m². The value for modulus of elasticity is 225 GPa.

$$a = \frac{2E\gamma_s}{\pi\sigma_c^2} = \frac{(2)(225 \times 10^9 \text{ N/m}^2)(1.0 \text{ N/m})}{(\pi)(13.5 \times 10^6 \text{ N/m}^2)^2}$$
$$= 7.9 \times 10^{-4} \text{ m} = 0.79 \text{ mm}$$

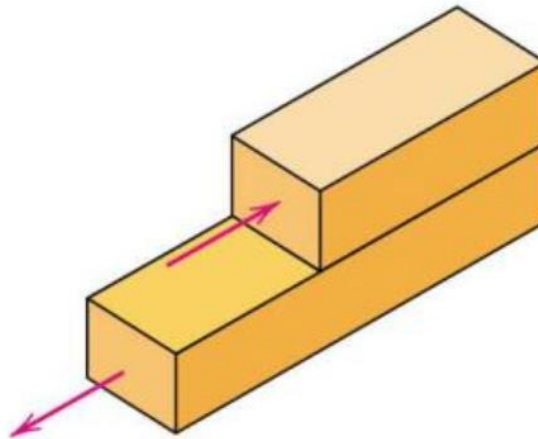
Principles of fracture mechanics

Typical crack propagation modes

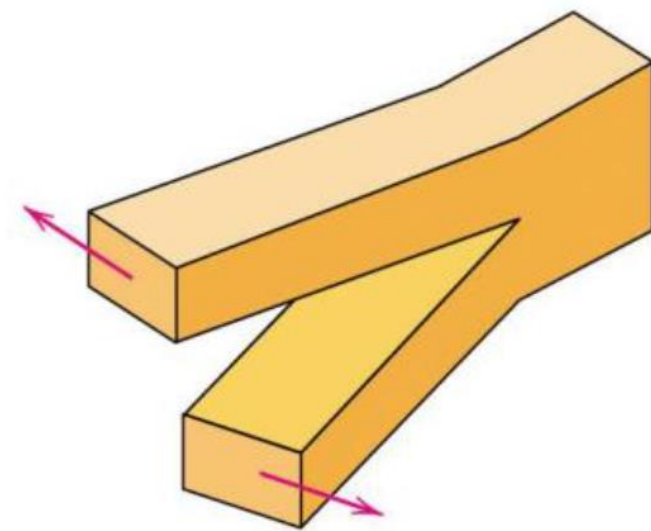
There are three main types of crack propagation modes.



Mode I
Opening or
Tensile mode



Mode II
Sliding mode



Mode III
Tearing mode

Principles of fracture mechanics

Stress intensity factor

- The stress intensity factor (K) is used to predict the stress intensity near a crack tip of a crack or notch tip caused by an applied remote load or residual stresses.
- Relevant expressions are derived for homogeneous and linear elastic materials (brittle materials or with small-scale yielding).
- The magnitude of K depends on the applied stress, geometry of the sample, the size and the position of the crack or notch as well as on their geometry.
- The general expression of K, derived from the solution of the elasticity equations is written as:

$$K = Y\sigma_0\sqrt{\pi a}$$

- Y: shape factor dependent on crack and specimen sizes and geometries as well as on loading mode;
- σ_0 : applied remote load.

Principles of fracture mechanics

Stress intensity factor

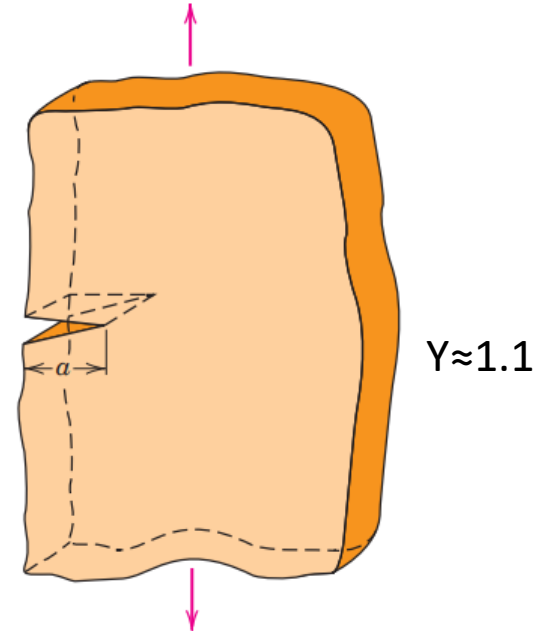
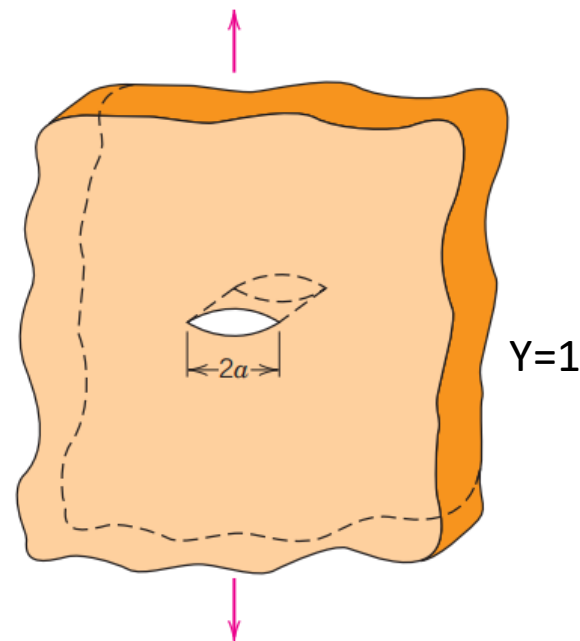
$$K = Y \sigma_0 \sqrt{\pi a}$$

- Y : shape factor dependent on crack and specimen sizes and geometries as well as on loading mode;
- σ_0 : applied remote load.

Mathematical expressions for Y have been determined for a variety of crack-specimen geometries. These expressions are rather easy in some cases, but for most cases are relatively complex.

For planar specimens containing cracks that are much shorter than the specimen width and for mode I, Y has a value of approximately unity.

$$K_I \approx \sigma_0 \sqrt{\pi a}$$



Poll – Answers in We Chat

Q1) Why do cracks occur in materials?

- A) Because of the presence of defects in crystals
- B) Because of the presence of impurities in crystals
- C) Because of the stress around defects in crystals

Poll – Answer

Q1) Why do cracks occur in materials?

C) Because of the stress around defects in crystals

Principles of fracture mechanics

Fracture toughness – mode I

Fracture Toughness is an indication of the amount of stress required to propagate a pre-existing flaw. In a specific mode of propagation, it is a material's property. It is defined as the highest value of stress intensity factor K_I that a material under very specific conditions can withstand without crack propagation. For mode I:

$$K_{IC} = Y \sigma_C \sqrt{\pi a}$$

As the stress intensity factor reaches the K_{IC} value, unstable fracture occurs.

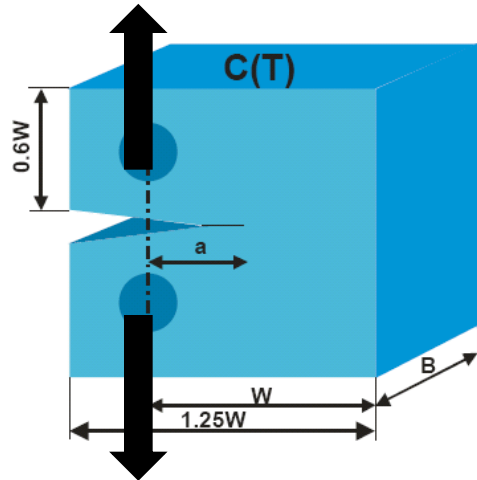
For relatively thin specimens, the value of K_C depends on specimen thickness. However, when specimen thickness is much greater than the crack dimensions, K_C becomes independent of thickness. These conditions are known as **plane strain conditions**.

Principles of fracture mechanics

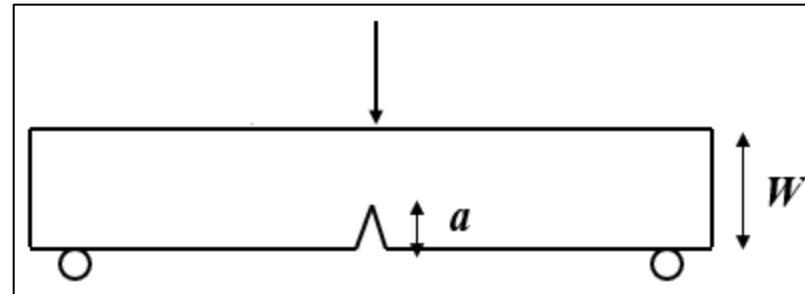
Fracture toughness testing

A number of different standardized tests have been devised to measure the fracture toughness values for structural materials. In these types of tests, the specimen (of specified geometry and size) contains a preexisting defect, usually a sharp crack. These tests include:

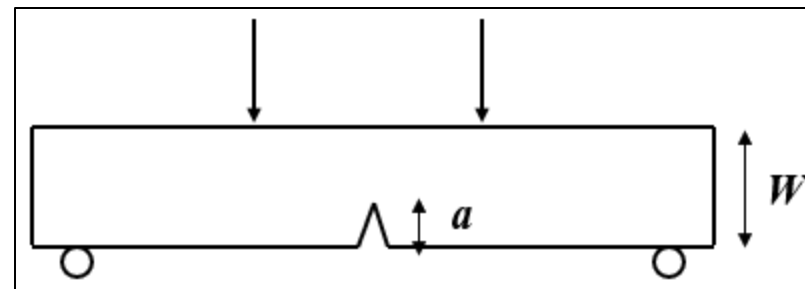
- Compact tension



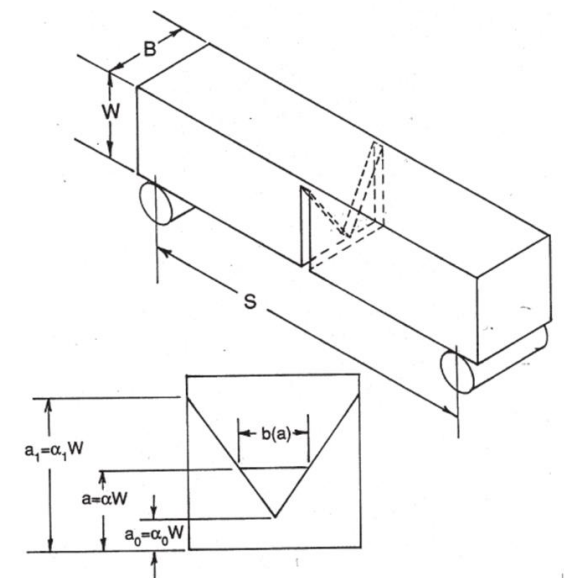
- Three-point bending



- Four-point bending



- Chevron notch

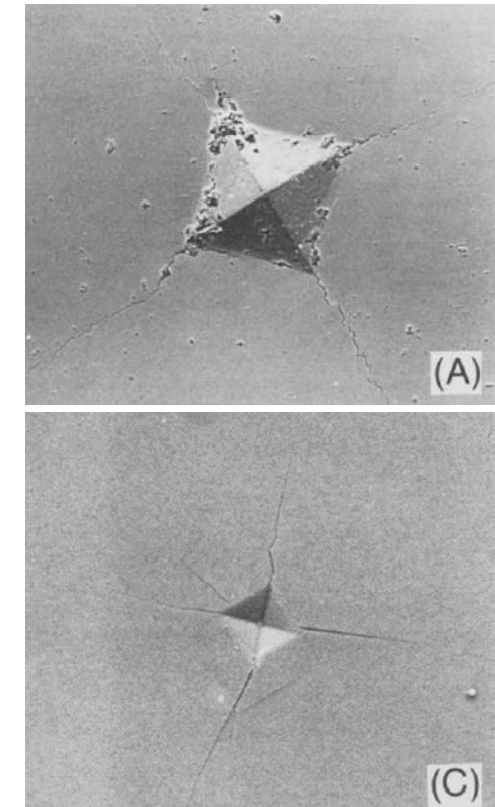
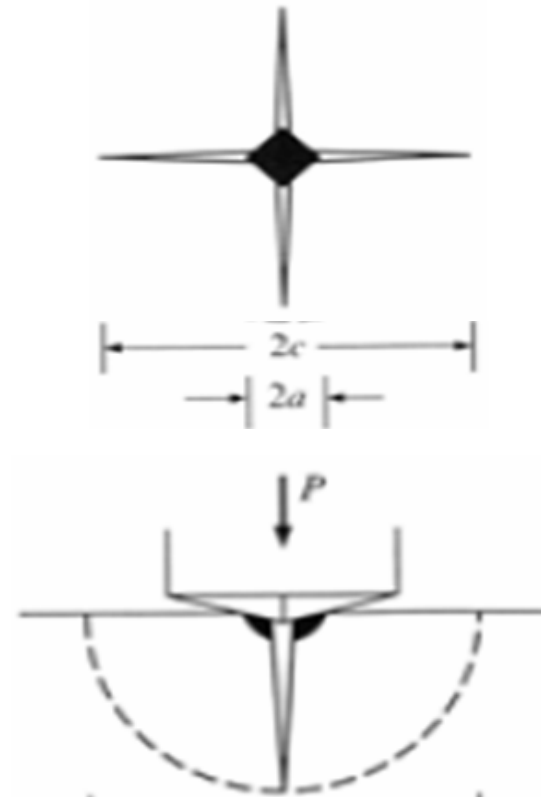
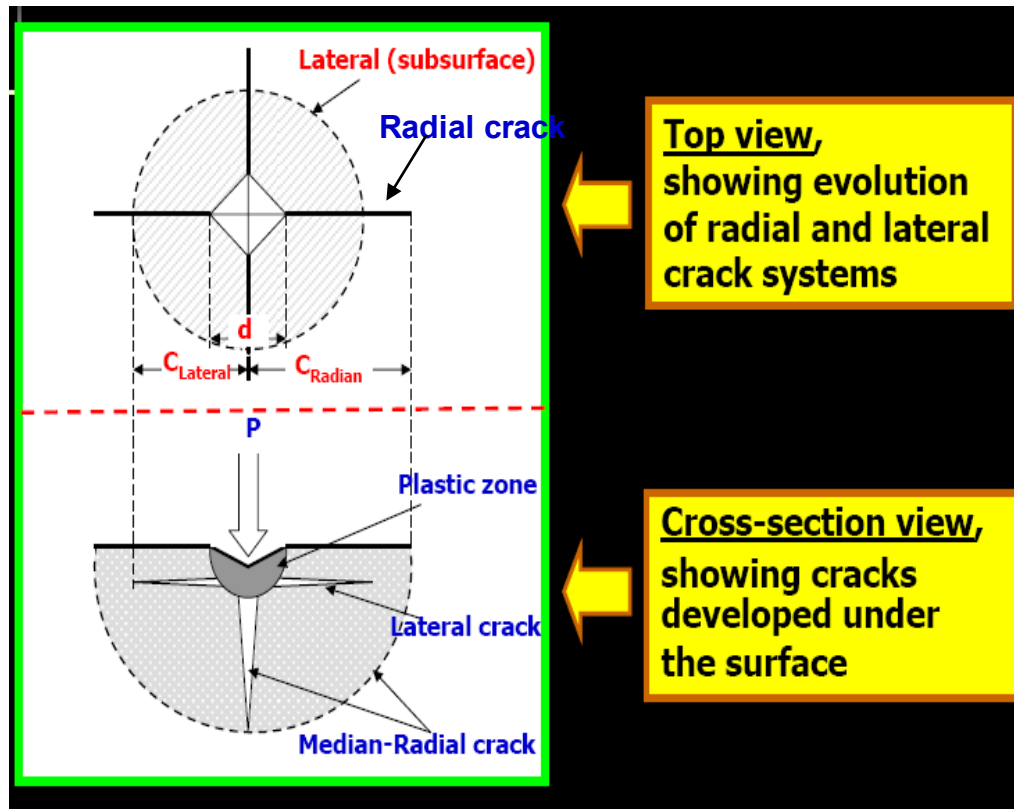


Principles of fracture mechanics

Fracture toughness testing

Indentation fracture toughness (non-standard method)

This method consists of estimating the fracture toughness from the length of indentation cracks developed in brittle materials from indentation tips. Although it is a non-standardized tests, it is widely used for ceramics, as it provides useful comparative estimations.



Principles of fracture mechanics

Fracture toughness comparison

<i>Material</i>	<i>Yield Strength</i>		<i>K_{Ic}</i>	
	<i>MPa</i>	<i>ksi</i>	<i>MPa √m</i>	<i>ksi √in.</i>
<i>Metals</i>				
Aluminum alloy ^a (7075-T651)	495	72	24	22
Aluminum alloy ^a (2024-T3)	345	50	44	40
Titanium alloy ^a (Ti-6Al-4V)	910	132	55	50
Alloy steel ^a (4340 tempered @ 260°C)	1640	238	50.0	45.8
Alloy steel ^a (4340 tempered @ 425°C)	1420	206	87.4	80.0
<i>Ceramics</i>				
Concrete	—	—	0.2–1.4	0.18–1.27
Soda-lime glass	—	—	0.7–0.8	0.64–0.73
Aluminum oxide	—	—	2.7–5.0	2.5–4.6
<i>Polymers</i>				
Polystyrene (PS)	25.0–69.0	3.63–10.0	0.7–1.1	0.64–1.0
Poly(methyl methacrylate) (PMMA)	53.8–73.1	7.8–10.6	0.7–1.6	0.64–1.5
Polycarbonate (PC)	62.1	9.0	2.2	2.0

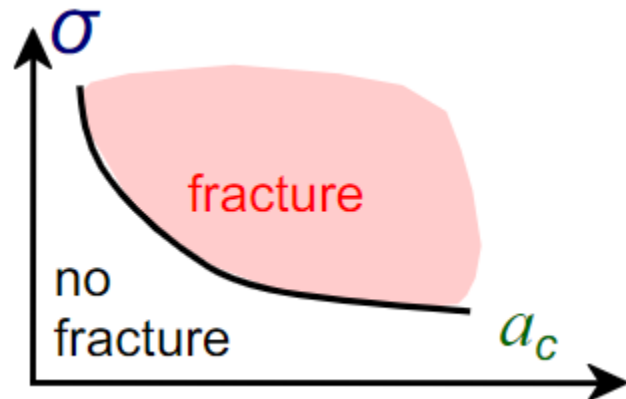
^a**Source:** Reprinted with permission, *Advanced Materials and Processes*, ASM International, © 1990.

Design using fracture mechanics

When designing a component, it is first important to decide which of these variables are constrained by the application and which are subject to design control.

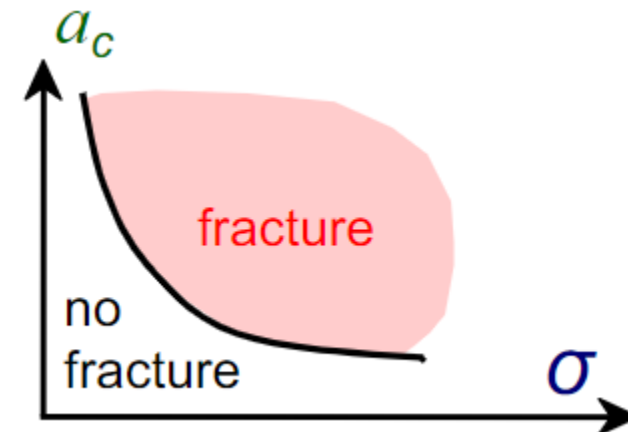
Scenario 1: Maximum flaw size dictates design stress.

$$\sigma_c < \frac{K_{1c}}{Y \sqrt{\pi a_c}}$$



Scenario 2: Design stress dictates maximum flaw size.

$$a_c < \frac{1}{\pi} \left(\frac{K_{1c}}{Y \sigma_c} \right)^2$$



Design using fracture mechanics

Example 5:

A material has a fracture toughness of $K_{IC} = 26 \text{ MPa}\sqrt{\text{m}}$. In a specific design, the failure stress in presence of a largest flaw of 9 mm is 112 MPa. Determine the failure stress in presence of a crack of 4 mm length.

Design using fracture mechanics

Example 5 solution:

A material has a fracture toughness of $K_{IC} = 26 \text{ MPa}\sqrt{\text{m}}$. In a specific design, the failure stress in presence of a largest flaw of 9 mm is 112 MPa. Determine the failure stress in presence of a crack of 4 mm length.

$$\sigma_c = \frac{K_{IC}}{Y\sqrt{\pi a_c}}$$

Y and K_{IC} are the same for both designs.

$$\frac{K_{IC}}{Y\sqrt{\pi}} = \sigma\sqrt{a} = \text{constant}$$

Therefore:

$$\sigma_{C1}\sqrt{a_{C1}} = \sigma_{C2}\sqrt{a_{C2}} \quad \Rightarrow \quad \sigma_{C2} = 168 \text{ MPa}$$

Design using fracture mechanics

Example 6:

A structural component in the form of a wide plate is to be fabricated from a steel alloy that has a plane-strain fracture toughness of $98.9 \text{ MPa}\sqrt{\text{m}}$ and a yield strength of 860 MPa . The flaw size resolution limit of the flaw detection apparatus is 3.0 mm . If the design stress is one-half the yield strength and the value of Y is 1.0 , determine whether a critical flaw for this plate is subject to detection.

Design using fracture mechanics

Example 6 solution:

A structural component in the form of a wide plate is to be fabricated from a steel alloy that has a plane-strain fracture toughness of 98.9 MPa $\sqrt{\text{m}}$ and a yield strength of 860 MPa. The flaw size resolution limit of the flaw detection apparatus is 3.0 mm. If the design stress is one-half the yield strength and the value of Y is 1.0, determine whether a critical flaw for this plate is subject to detection.

$$a_c = \frac{1}{\pi} \left(\frac{K_{Ic}}{Y\sigma_c} \right)^2 = \frac{1}{\pi} \left[\frac{98.9 \text{ MPa}\sqrt{\text{m}}}{(1.0) \left(\frac{860 \text{ MPa}}{2} \right)} \right]^2$$
$$= 0.0168 \text{ m} = 16.8 \text{ mm}$$

Therefore, the critical flaw *is subject to detection* since this value of a_c (16.8 mm) is greater than the 3.0 mm resolution limit.

Design using fracture mechanics

Example 7:

Calculate the maximum internal crack length allowed for a Ti-6Al-4V titanium alloy component that is loaded to a stress one-half its yield strength. Assume that the value of Y is 1.50.

$K_{IC} = 55\text{MPa}\sqrt{\text{m}}$ and $\sigma = 455\text{MPa}$.

Design using fracture mechanics

Example 7 solution:

Calculate the maximum internal crack length allowed for a Ti-6Al-4V titanium alloy component that is loaded to a stress one-half its yield strength. Assume that the value of Y is 1.50.

$K_{IC} = 55 \text{ MPa} \sqrt{\text{m}}$ and $\sigma = 455 \text{ MPa}$.

Since this is an internal crack we have to solve for $2a_c$

$$2a_c = \frac{2}{\pi} \left(\frac{K_{Ic}}{Y\sigma} \right)^2 = \frac{2}{\pi} \left[\frac{55 \text{ MPa} \sqrt{\text{m}}}{(1.5)(455 \text{ MPa})} \right]^2 = 0.0041 \text{ m} = 4.1 \text{ mm}$$

Summary

- Metals can fail due to cracks and voids internal to their structure.
- Fracture may occur by ductile and brittle modes.
- Ductile fracture requires more energy than brittle fracture.
- Cracks in ductile materials are considered stable, while unstable in brittle materials, in which cracks propagate much faster.
- For ductile metals, two tensile fracture profiles are possible: i) large necking when ductility is high; ii) cup-and-cone fracture profile in less-ductile materials.
- In brittle materials, the fracture surface is relatively flat and perpendicular to the direction of the applied tensile load. Transgranular and intergranular crack propagation can occur.
- The discrepancy between actual and theoretical fracture strengths is due to defects that amplify the applied tensile stress in their vicinity, as quantified by the stress intensity factor.
- The fracture toughness of a material is indicative of its resistance to brittle fracture when a crack is present.

General learning outcomes of the entire unit on failure

Learning
objective

&

Learning
outcomes

Introduce

Introduce the concepts of fracture, fatigue and creep

Differentiate

Differentiate between the mechanism of crack propagation for ductile and brittle modes of fracture.

Explain

Explain why the strengths of brittle materials are much lower than predicted by theoretical calculations

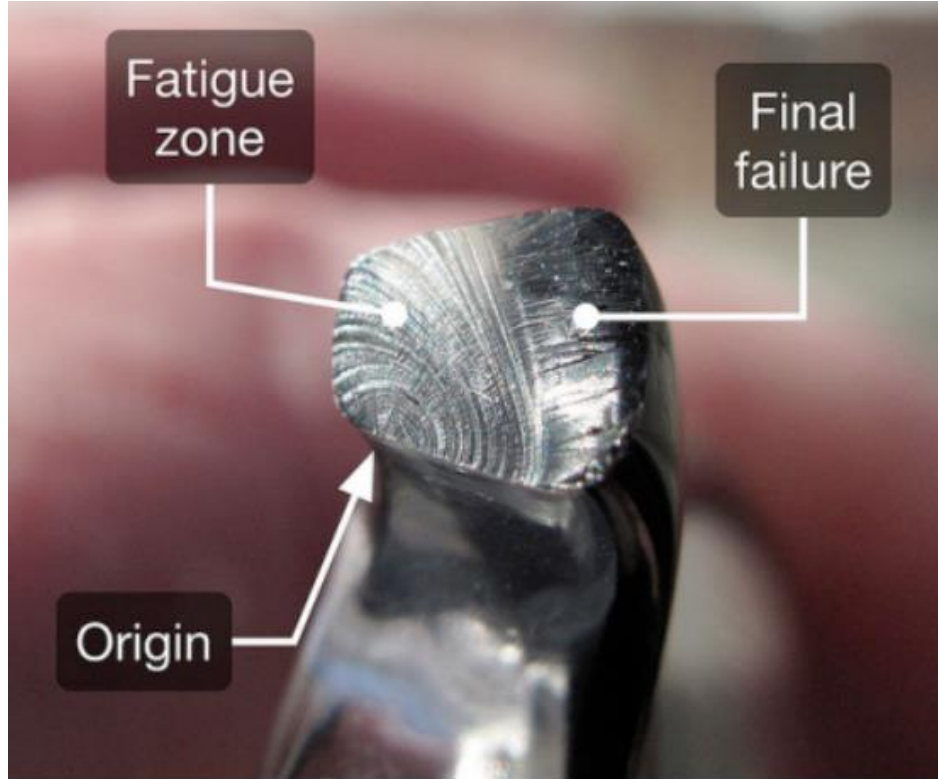
Describe

Describe the mechanism of fatigue and creep, including cyclic loading

Learning outcomes of this lecture

- Introduce the concepts of fatigue and creep.
- Distinguish failure by fatigue and creep.
- Use data from S-N curves to calculate fatigue life of materials.
- Calculate the deformation of materials due to creep.

Fatigue



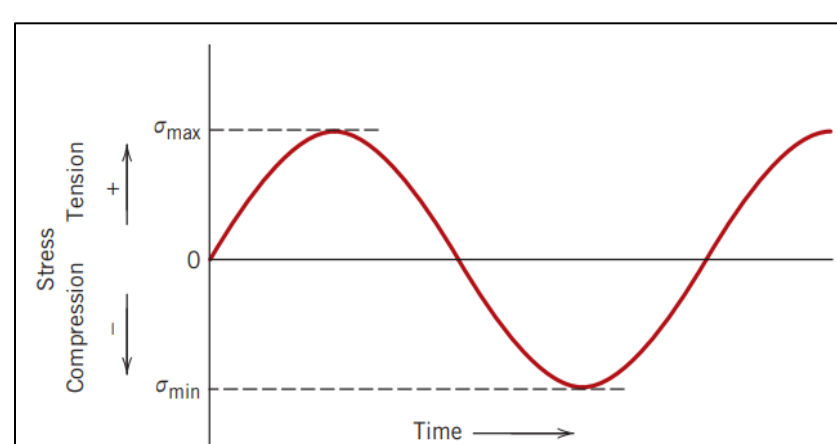
- Fatigue is a form of failure that occurs in structures subjected to dynamic and fluctuating stresses for long period.
- Fatigue failure can occur at lower stresses than the tensile or yield strength for a static load.
- Fatigue failure is brittle-like in nature (sudden and catastrophic) even in normally ductile metals. The process occurs by the initiation and propagation of cracks, and typically the fracture surface is perpendicular to the direction of an applied tensile stress.
- Fatigue is the single largest cause of failure in metals (90%). Polymers and ceramics (except for glasses) can also fail by fatigue.

Fatigue

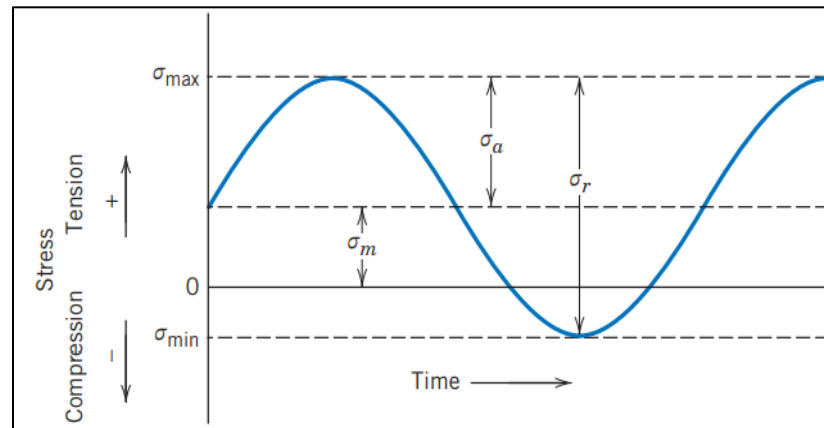
Cyclic stresses

The applied stress may be axial (tension–compression), flexural (bending), or torsional (twisting) in nature.

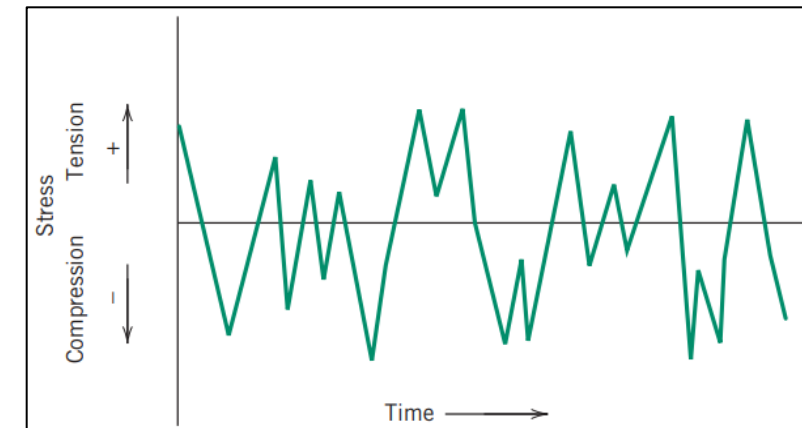
In general, three different fluctuating stress–time modes are possible.



Regular and sinusoidal time dependence. The stress amplitude is symmetrical about a mean of zero stress, alternating from a maximum tensile stress ($\sigma_{\max}$) to a minimum compressive stress ($\sigma_{\min}$) of equal magnitude.



The maxima and minima are asymmetrical relative to the zero-stress level.



Stress level varies randomly in amplitude and frequency.

Fatigue

S-N curves

- The stress (S) vs number of cycles (N) curves are generated by subjecting a specimen to stress cycling at a relatively large maximum stress ($\sigma_{\max}$), usually on the order of two-thirds of the static tensile strength. The number of cycles to failure is counted and recorded.
- The procedure is repeated on other specimens at progressively decreasing maximum stress levels.
- Data are plotted as stress S versus the logarithm of the number N of cycles to failure for each of the specimens.
- The S parameter is normally taken as either maximum stress ($\sigma_{\max}$) or stress amplitude (σ_a).

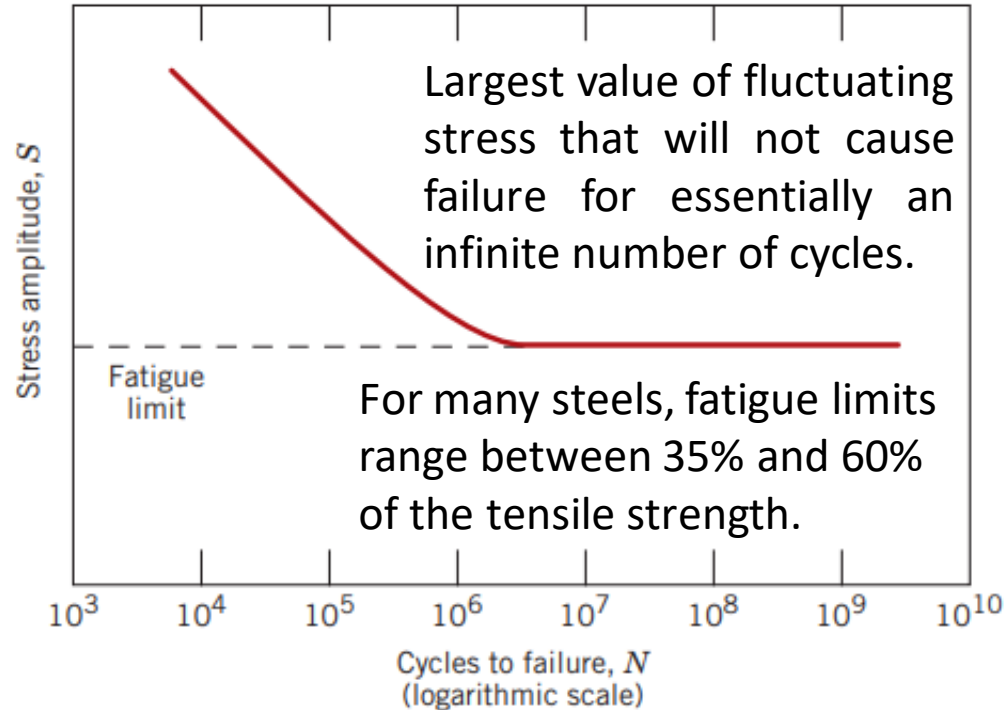
Fatigue

S-N curves

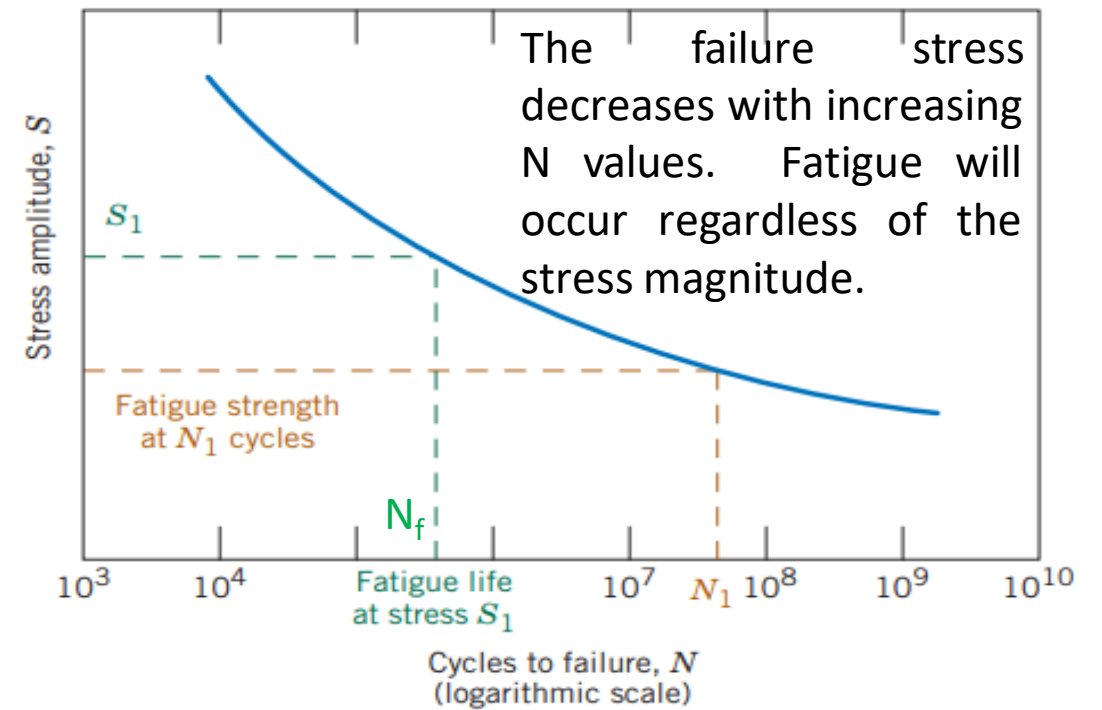
The S-N curves indicate that the higher the magnitude of the stress, the smaller the number of cycles the material is capable of sustaining before failure.

Two types of S–N behaviour can be distinguished:

1. S-N curves with a fatigue limit (iron- and titanium-based alloys).



2. S-N curves without fatigue limit (most non-ferrous alloys).



Fatigue

Example 1:

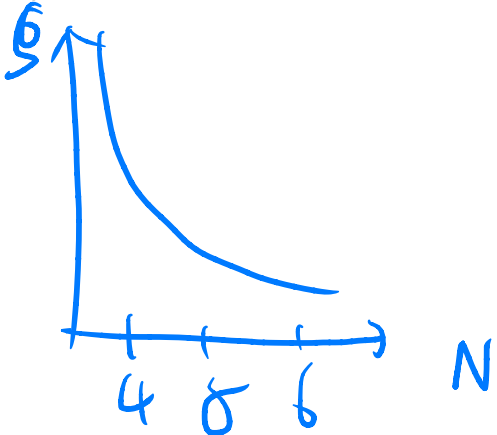
黄铜合金

The fatigue data for a brass alloy are given in the table.

Using this data:

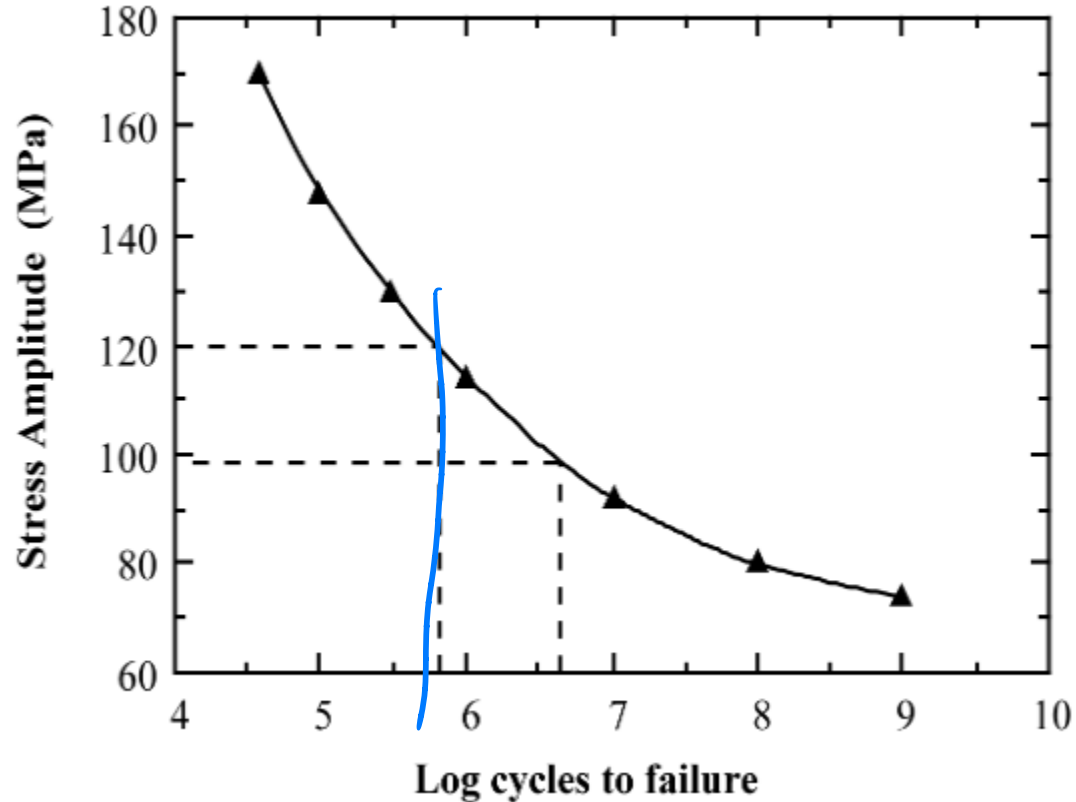
- (a) Make an S–N plot (stress amplitude versus logarithm of cycles to failure);
- (b) Determine the fatigue strength at 4×10^6 cycles;
- (c) Determine the fatigue life for 120 MPa.

<i>Stress Amplitude (MPa)</i>	<i>Cycles to Failure</i>
170	3.7×10^4
148	1.0×10^5
130	3.0×10^5
114	1.0×10^6
92	1.0×10^7
80	1.0×10^8
74	1.0×10^9



Fatigue

Example 1 solution:



The fatigue strength @ 4×10^6 cycles [$\log(4 \times 10^6) = 6.6$] is ~ 100 MPa

在图里画

The fatigue life for 120 MPa is $\sim 6 \times 10^5$ cycles (i.e., the log of the lifetime is ~ 5.8)

Fatigue

Factors affecting fatigue

There are many factors affecting fatigue. These include:

- Mean stress level;
- Surface effects (presence of surface cracks and residual stresses);
- Environmental effects (temperature, corrosive environment).

Fatigue

Fatigue failure process

The process of fatigue failure is characterized by three distinct steps:

- (1) crack initiation, in which a small crack forms at some point of high stress concentration;
- (2) crack propagation, during which this crack advances incrementally with each stress cycle;
- (3) final failure, which occurs very rapidly once the advancing crack has reached a critical size.

→ 逐渐发展

Cracks associated with fatigue failure almost always initiate on the surface of a component at some point of stress concentration, such as surface scratches, threads, dents, and similar.

In addition, cyclic loading can produce microscopic surface discontinuities resulting from dislocation slip steps that may also act as stress raisers and therefore as crack initiation sites.

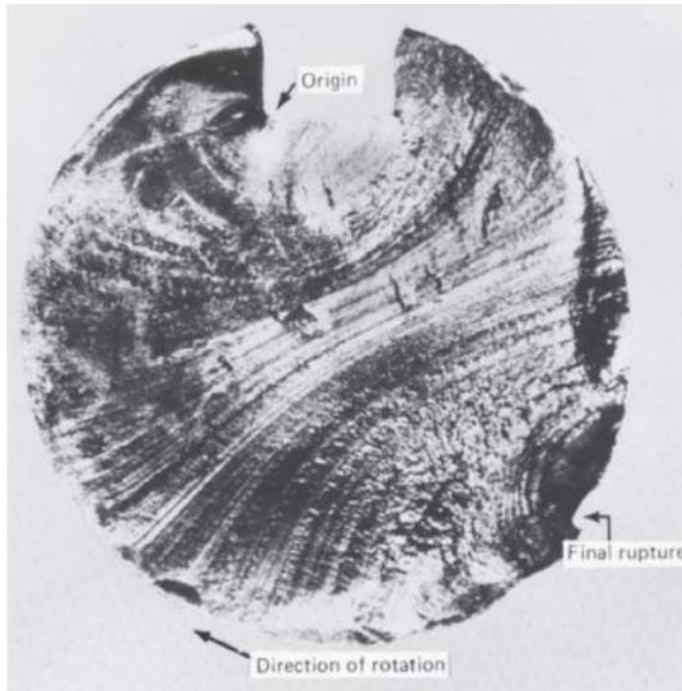
Fatigue

Fractography of fatigue failure

The region of a fracture surface that formed during the crack propagation step may be characterized by two types of markings termed **beach-marks and striations**.

Both appear as concentric ridges that expand away from the crack initiation site, frequently in a circular or semicircular pattern.

Beachmarks



Beachmarks (sometimes also called clamshell marks) are of macroscopic dimensions and may be observed by naked eye in components that experienced interruptions during the crack propagation (i.e. machine operated only during normal workshift hours).

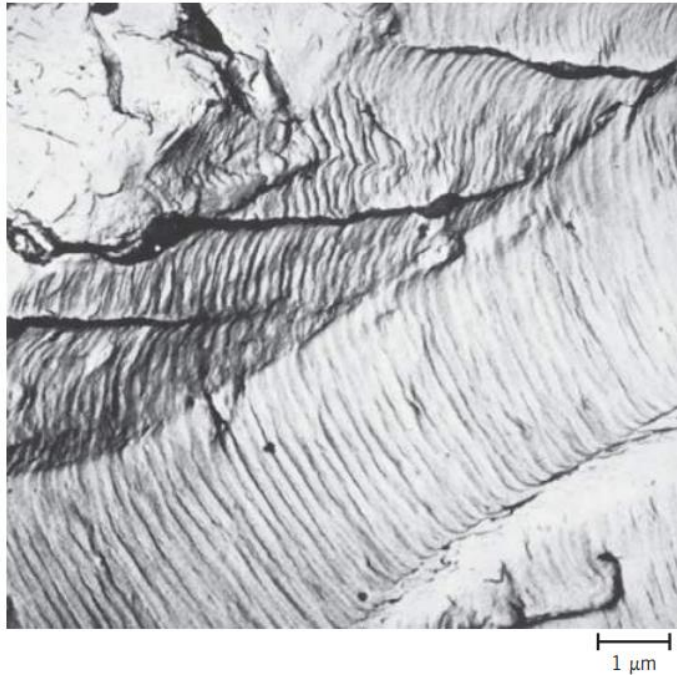
Each beachmark band represents a period over which crack growth occurred.

Fatigue

Fractography of fatigue failure

Fatigue striations are microscopic in size and visible by electron microscopy (either TEM or SEM).

Striations



Each striation may represent the advance distance of a crack front during a single load cycle.

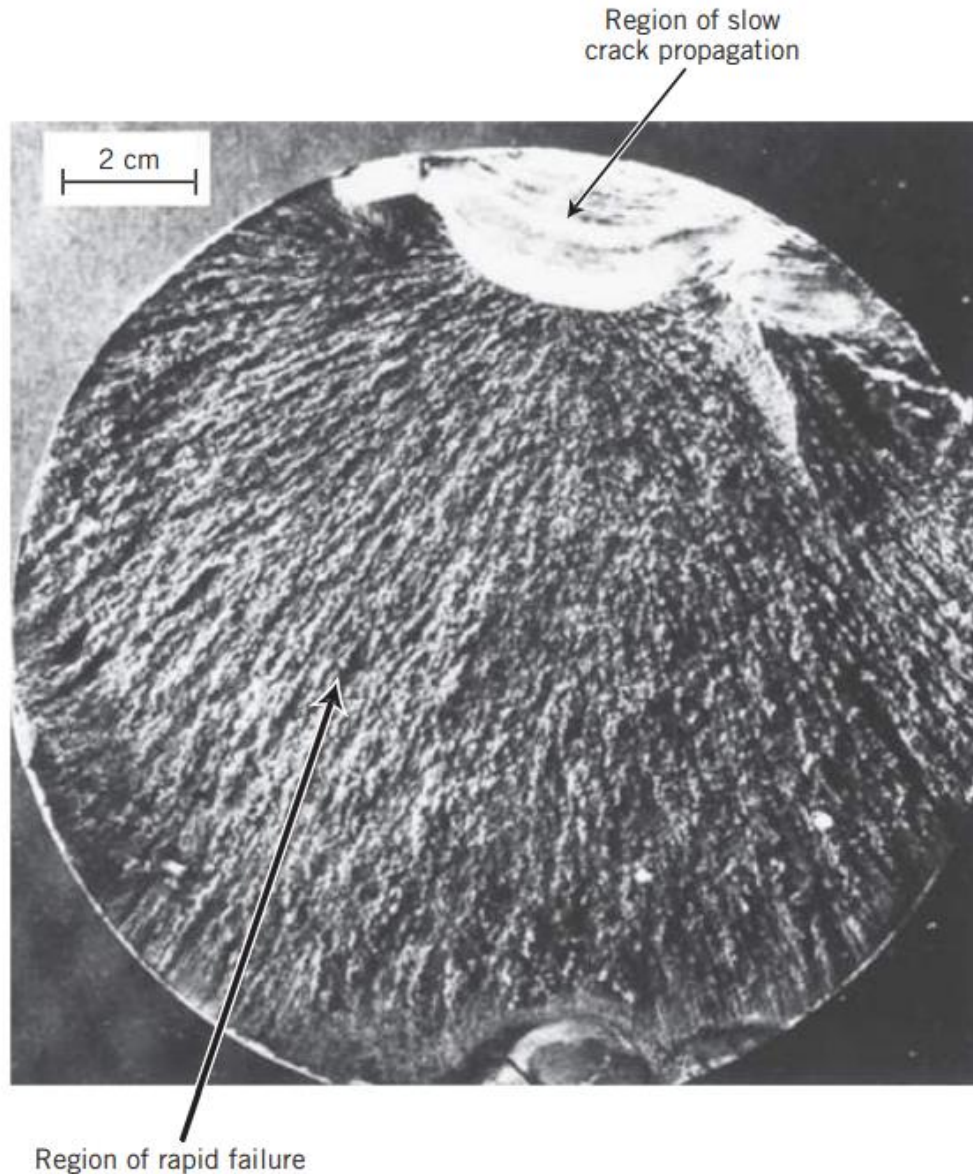
Striation width increases with increasing stress range. The applied stress is amplified at crack tips to the degree that local stress levels exceed the yield strength.

The geometry of fatigue striations is a manifestation of this plastic deformation.

Beachmarks and striations have similar appearances, but they are different in both origin and size. There may be thousands of striations within a single beachmark.

Fatigue

Fractography of fatigue failure



Beachmarks and striations do not appear on the region over which the rapid failure occurs.

The smooth region near the top corresponds to the area over which the crack propagated slowly.

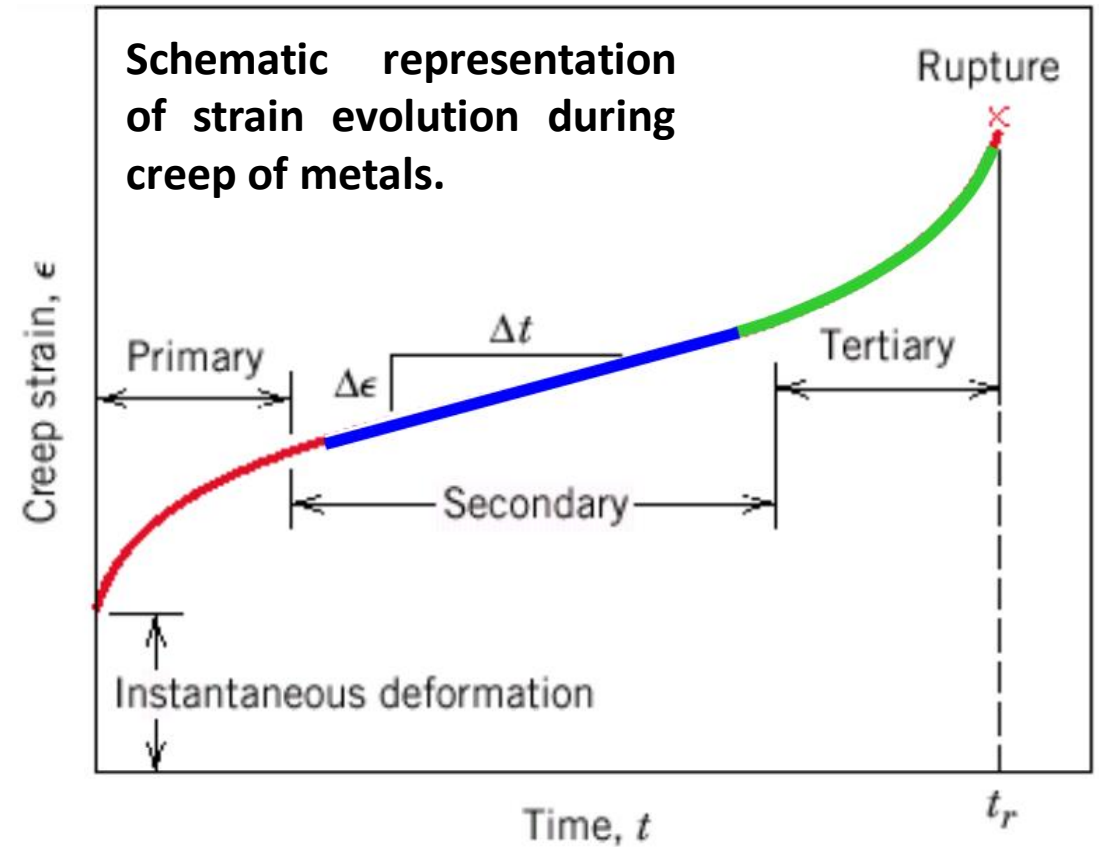
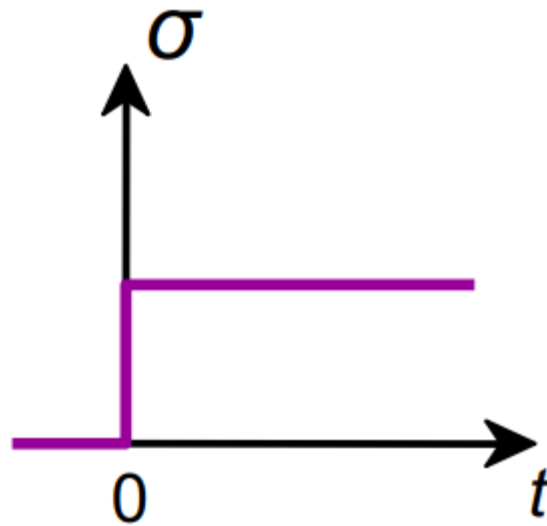
Rapid failure occurred over the area having a dull and fibrous texture (the largest area).

Creep

- Creep is the phenomenon by which a sample experiences time-dependent and permanent deformations under a constant stress (σ).
- Creep is usually normally an undesirable and is often the limiting factor in the lifetime of engineering components.
- It is observed in all materials types; for metals, it becomes important only for temperatures greater than about $0.4T_m$, where T_m is the absolute melting temperature.
- However, there are materials that experience time-dependent deformations also at room temperature, such as viscoelastic and ferroelastic materials.

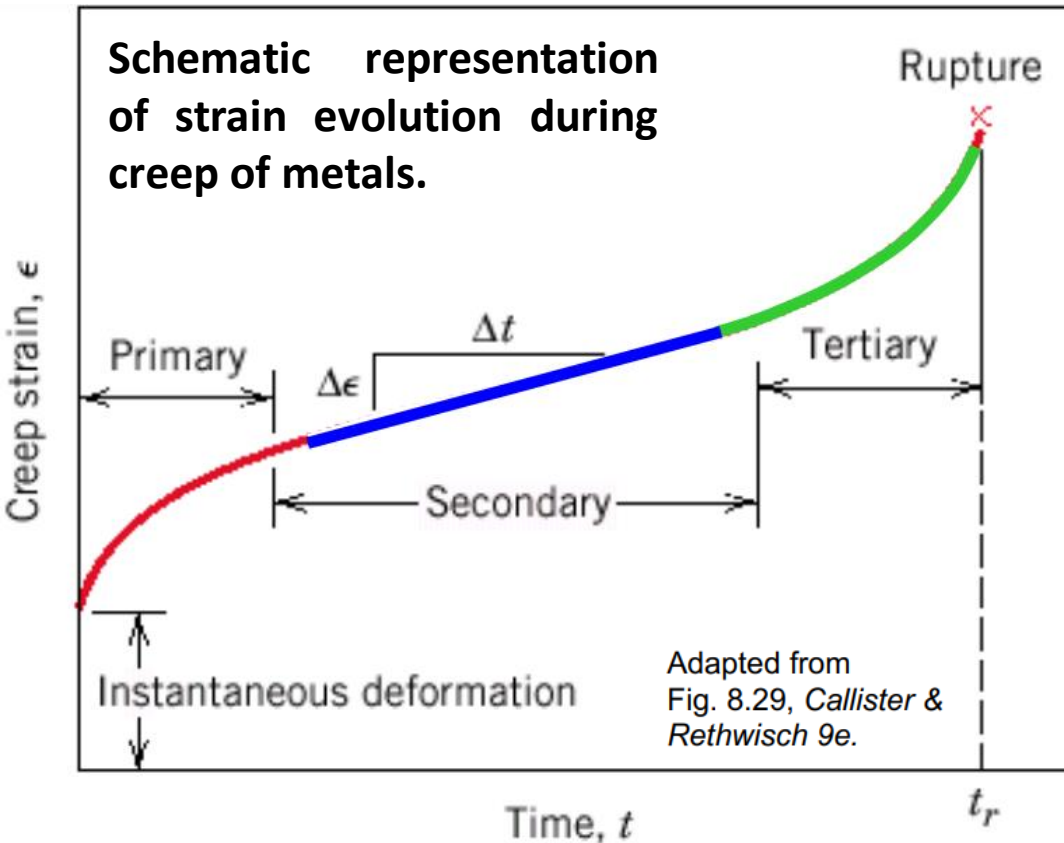
Creep

Typical creep test: apply a constant load or stress, while maintaining the temperature constant. The deformation or strain is measured and plotted as a function of time.



Creep

Strain evolution during creep: there are different stages that can be identified based on the strain-time diagram.



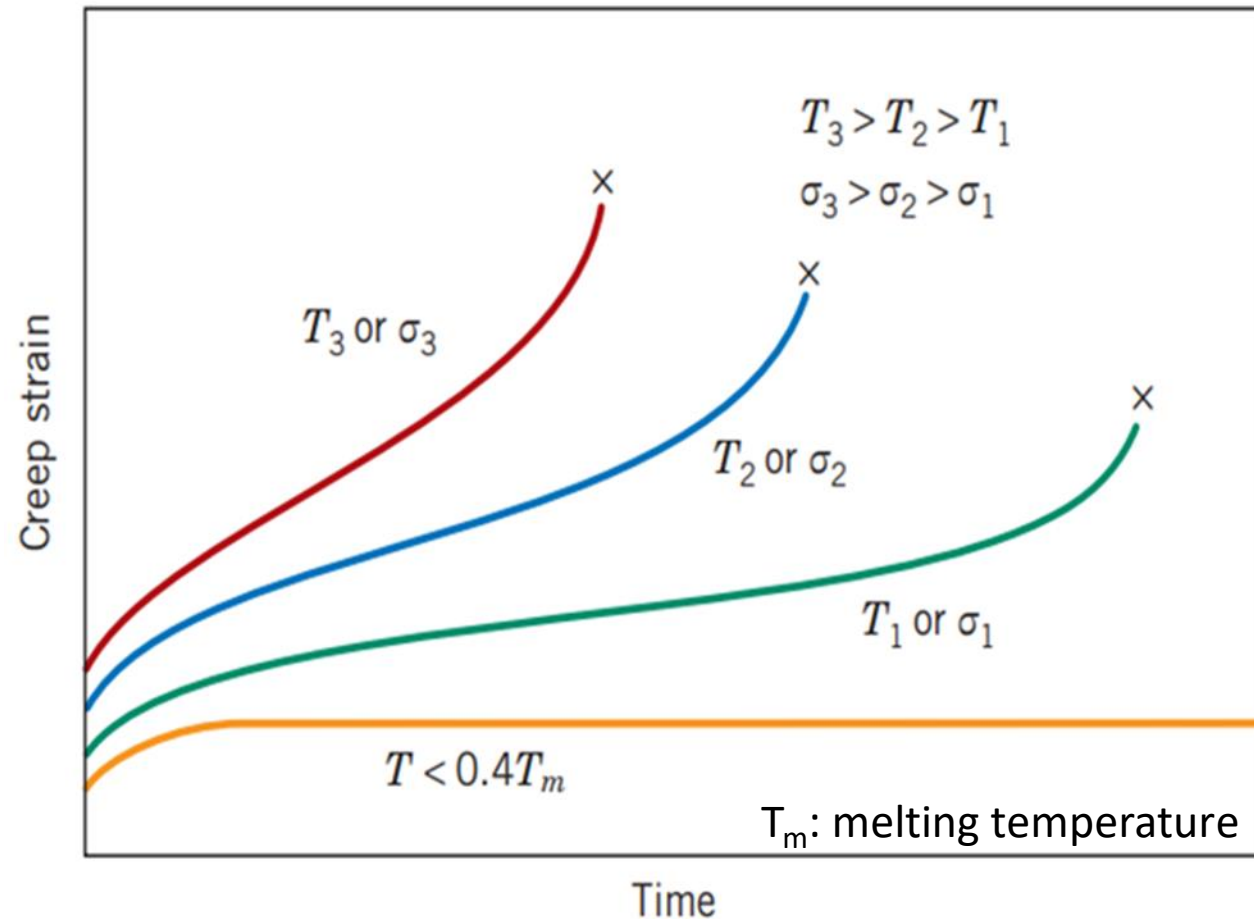
Upon application of the load, there is an instantaneous deformation, which is totally elastic.

Creep is characterized by 3 main stages:

1. **Primary or transient creep:** strain rate (slope) decreases with time. The material experiences creep resistance or strain hardening, making deformation more difficult.
2. **Secondary creep or steady-state creep (stage of longest duration):** strain rate is constant (linear plot). This is due to a balance between strain hardening and recovery.
3. **Tertiary creep:** strain rate increases towards rupture, which results from grain boundary separation, formation of internal cracks, cavities, and voids.

Creep

Effect of stress and temperature: with increasing temperature and stress, instantaneous strain and strain rate in all creep stages increase, while the time to achieve rupture or failure decreases.



The dependence of creep strain on stress and temperature is described using the following relationships:

$$\dot{\epsilon}_s = K_1 \sigma^n$$

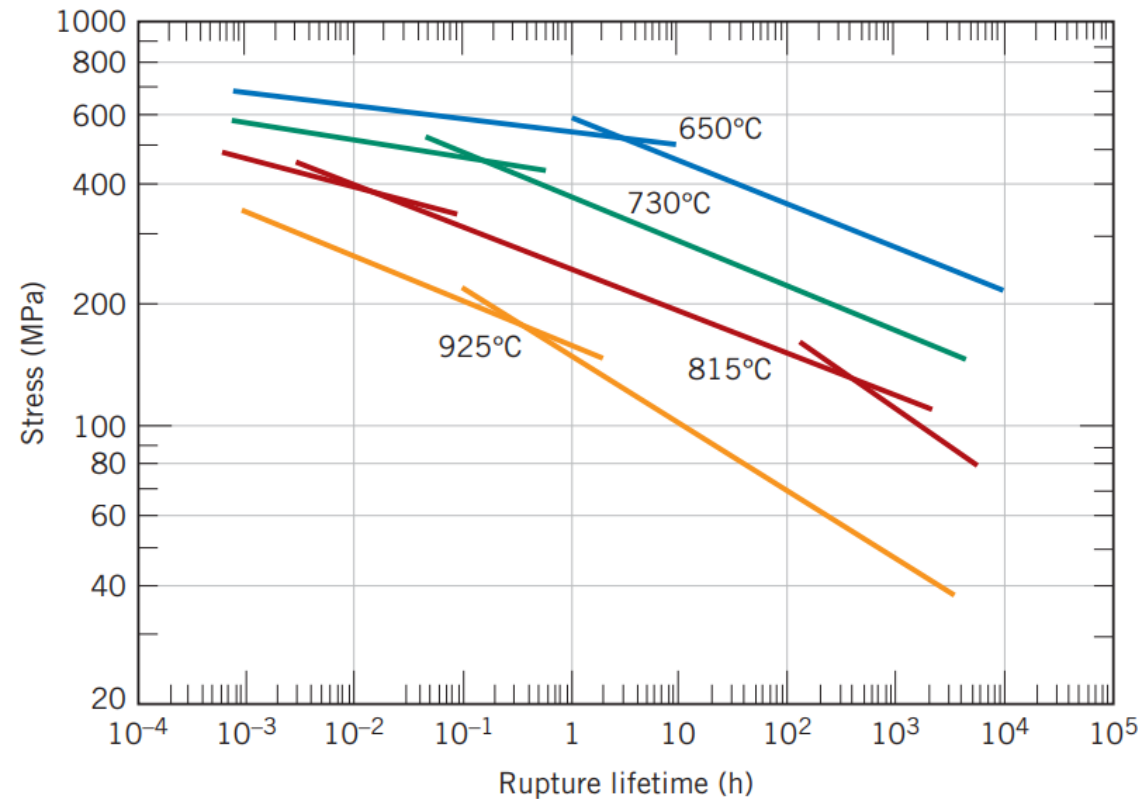
$$\dot{\epsilon}_s = K_2 \sigma^n \exp\left(-\frac{Q_c}{RT}\right)$$

K_1 , K_2 , n are constants. R is the gas constant. Q_c is the activation energy of the creep process.

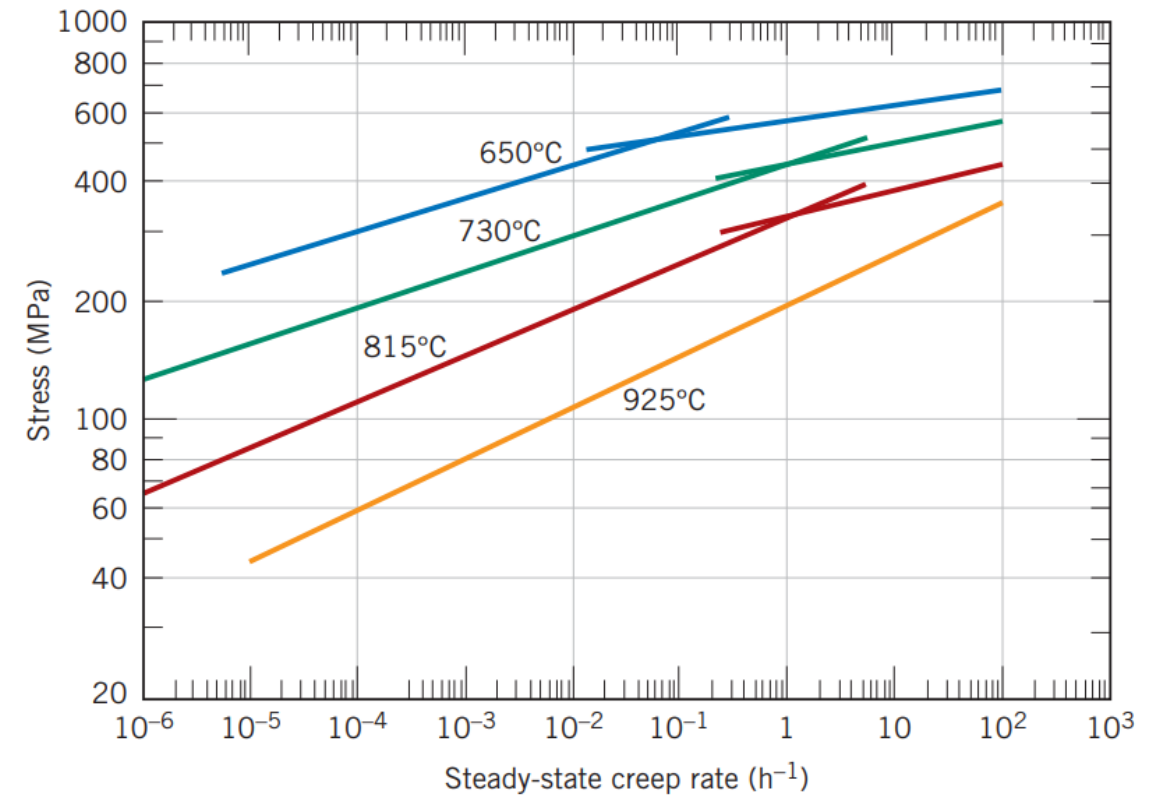
Creep

Effect of stress and temperature:

The plots of logarithm of stress versus the logarithm of rupture lifetime for an S-590 alloy appear as linear at different temperatures.



The plots of logarithm of stress against the logarithm of strain rate in steady state for an S-590 alloy appear as linear at different temperatures.



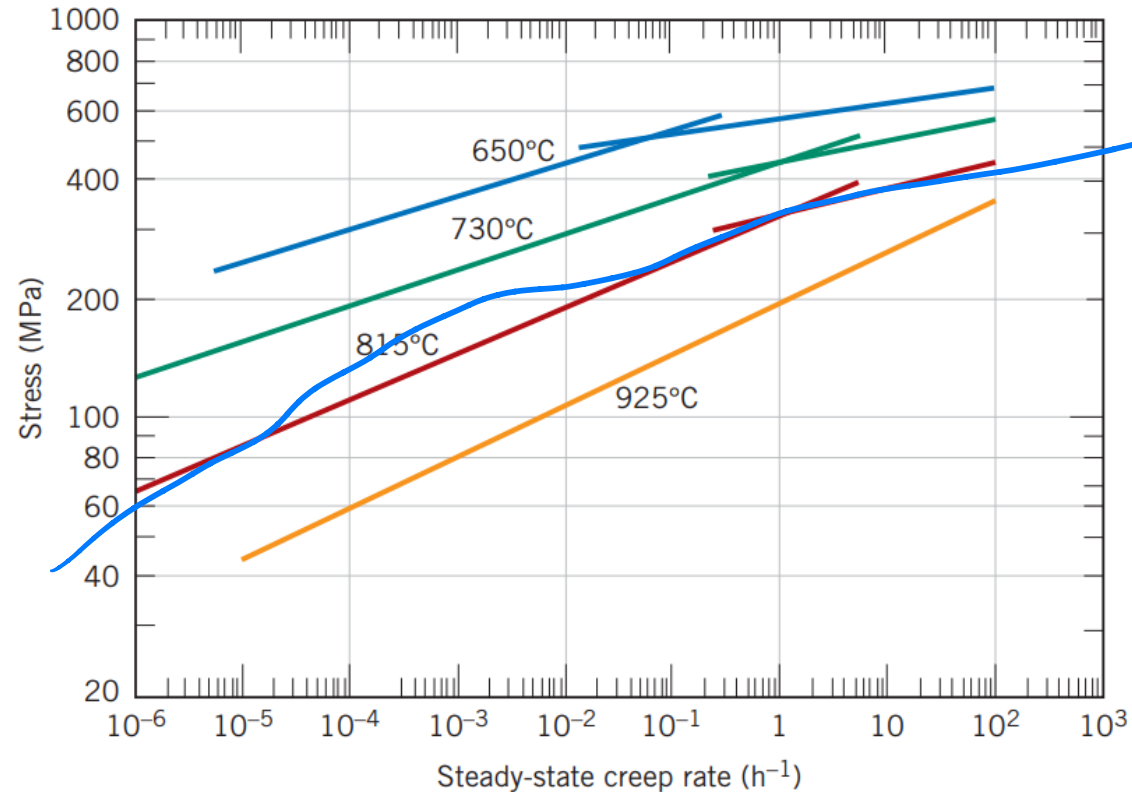
Creep

Example 2:

A 750 mm specimen of S5-90 alloy is exposed to a tensile stress of 80 MPa at 815 °C. Determine its elongation after 5000 h, assuming an instantaneous and primary creep elongation of 1.5 mm.

8/5+2/3

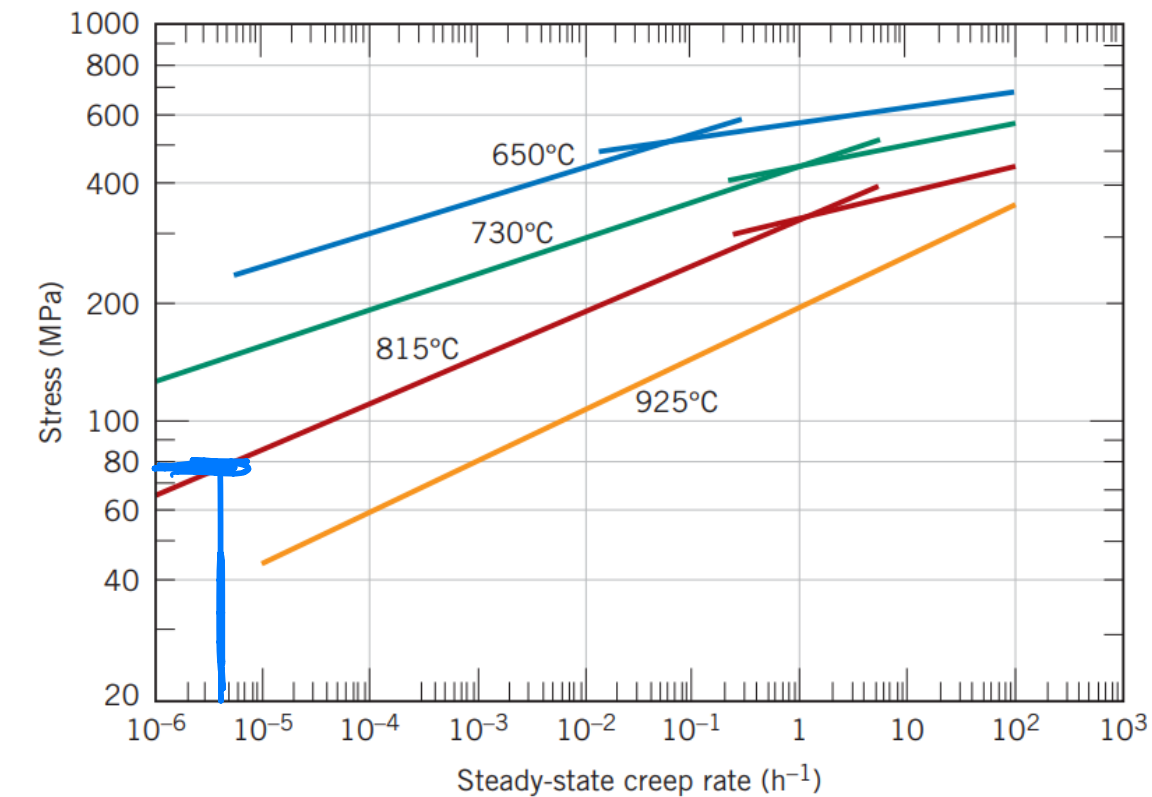
$\epsilon_s =$



Creep

Example 2 solution:

A 750 mm specimen of S5-90 alloy is exposed to a tensile stress of 80 MPa at 815°C. Determine its elongation after 5000 h, assuming an instantaneous and primary creep elongation of 1.5 mm.



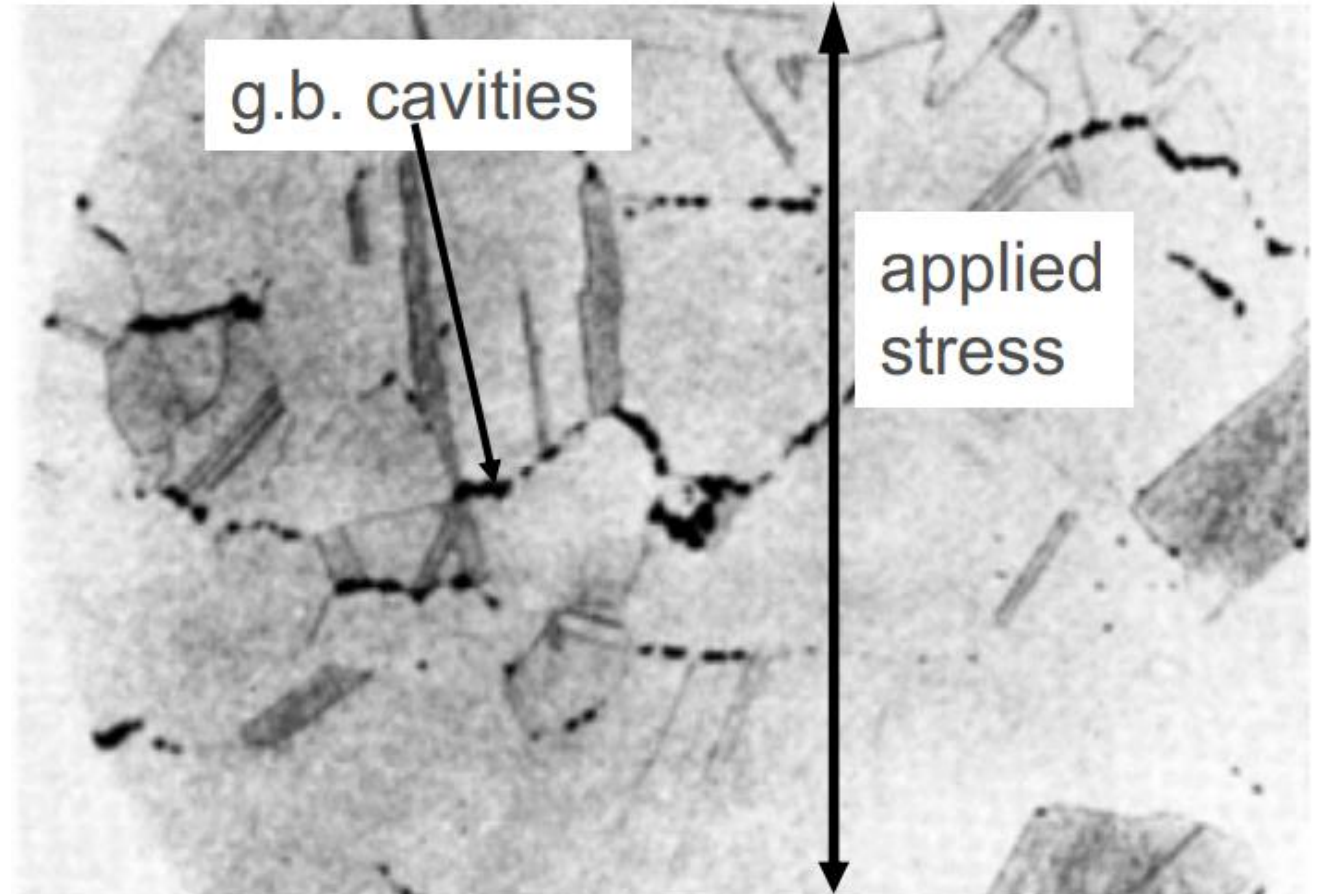
- Strain rate in the steady state can be obtained from the graph. At 80 MPa, the strain rate is $4 \times 10^{-6} \text{ h}^{-1}$.
- Multiplying the steady strain rate creep rate by time: $(5000\text{h})(4 \times 10^{-6} \text{ h}^{-1}) = 0.02$, one obtains the strain after 5000 h of steady state creep.
- The elongation can be obtained from the definition of strain: $\frac{\Delta L}{L_0} = 0.02$. It follows: $\Delta L_{ss} = 750 \times 0.02 = 15 \text{ mm}$.
- Therefore, the final elongation is obtained as:
$$\Delta L_{ss} + \Delta L_{ip} = 15 + 1.5 = 16.5 \text{ mm}$$

总长

Creep

Failure mechanisms

- Stress-induced vacancy diffusion
- Dislocation motion
- Grain boundary diffusion
- Grain boundary sliding
- Both stress and temperature affect creep behaviour



From V.J. Colangelo and F.A. Heiser, *Analysis of Metallurgical Failures* (2nd ed.), Fig. 4.32, p. 87, John Wiley and Sons, Inc., 1987. (Orig. source: Pergamon Press, Inc.)

Summary

- Fatigue is a common type of catastrophic failure in which the applied stress level fluctuates with time; it occurs when the maximum stress level may be considerably lower than the static tensile or yield strength.
- Stress is plotted against the logarithm of the number of cycles to failure (S-N curves). Two of S-N can be found in metal alloys: i) S-N curves with fatigue limit; ii) S-N curve without fatigue limit.
- Fatigue cracks normally nucleate on the surface of a component at some point of stress concentration. Fatigue fracture surface shows beachmarks and striations.
- Creep strain under constant stress at $T > 0.4T_m$ is characterized by three distinct regions: i) transient (or primary); ii) steady-state (or secondary); and iii) tertiary.
- Both temperature and applied stress level influence creep behaviour. Increasing either of these parameters enhances strain rate and accelerates failure.